TEAM INVICTUS

Systems Thinking & Design
Phase 3 and Phase 4 Submission

The Breakfast Paradox

IIIT Hyderabad Breakfast Mess System

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Phase 3: Leverage Points and Design Principles (Activity 9)

The Breakfast Paradox: IIIT Hyderabad Breakfast Mess System, Team Invictus

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Introduction

Phase 2 found that the Breakfast Paradox is a stable equilibrium, held in place by three confirmed structures: an automatic registration default with a narrow exit, billing at the booking and not at the meal, and a gap that the scan record and the operators' books both measure but that reaches nobody who sets the rules. On the kitchen side, registrations turn into a cooked first batch that any kitchen can top up and none can shrink. Activity 9 asks where this system can change. It delivers the Leverage-Point / System Intervention Map and the Design Principles, and hands seven principles and fourteen intervention points to Activity 10.

Claims keep the project's labels: confirmed (read off a record or seen on site), single-sourced (one account), candidate (an inference) and assumed. Depth is counted on Meadows' twelve places, with place 1 the deepest and place 12 the shallowest. Bakul has no records of any kind, so no Bakul effect is sized anywhere in this document.

1 Activity 9: Define Design Principles and Intervention Opportunities

Four methods run in turn: leverage points, read against archetypes and ranked; changes to rules, roles, incentives, information flows and processes; the design objective set against the case for the status quo; and fourteen intervention points. The map and the principles close the activity.

1.1 Identify Leverage Points

A lever is ruled out if it removes a mechanism that protects someone: resale, reuse of leftovers, the meals of 30 to 50 outside diners and about half of Kadamba's staff, the walk-in buffer, late and back-door service. The over-cooked first batch, repeated reheating and the bin protect no one and are fair targets. Feasibility is judged for a student team, which can propose, pilot and evidence a change but cannot enact a rule. High means a prototype or pilot with an actor already engaged, medium an evidenced proposal to an existing forum, low a proposal only, either because the holder is out of reach or because the lever rests on a fact nobody knows yet.

Rules (place 5) are used only where a rule itself changes. Place 9, on delays, would mean flattening the closing crest, which no lever attempts. Places 11 and 12 appear only as settings inside a deeper lever: cutting Kadamba's 70% ratio by decree would move the cost from surplus into run-outs at the crest, and the cancellation cap, raised from five to ten during a gas shortage and back at five now (single-sourced), changes only how many exits the default allows.

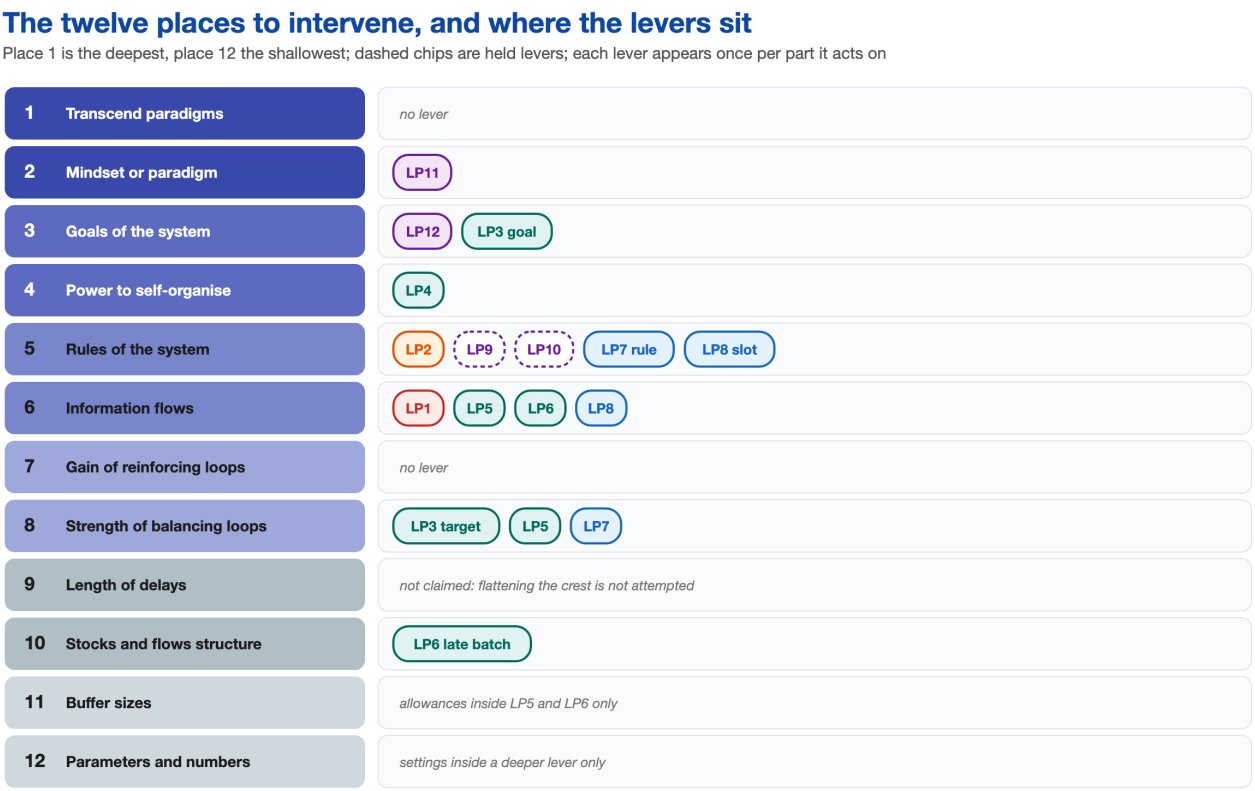


Figure 1. *The twelve places to intervene, and where each lever sits.*

Leverage point	Place	Holder of the power	Feasibility
LP1 Build the missing routes	6 (information flows)	CDS (scans), operators (books); receiver unnamed	High to prototype, medium to deliver
LP2 Change the breakfast default and its exit	5 (rules)	Registration rule owner (Mess Office or CDS, unconfirmed)	Medium
LP3 Operator goal: surplus is a cost	3 (goals); 8 for the target	Prism management; Bakul manager and firm	Low to medium
LP4 A standing operator forum that runs its own trials	4 (self-organisation)	Operators; CDS as convenor	Medium
LP5 Record-calibrated first batch with a daily learning record	6 and 8; allowances at 11 and 12	The operator	High (Yuktāhār), medium (Kadamba), unsizeable (Bakul)
LP6 Crest-aware top-up and a staged late batch	6 and 10	Operators; Mess Office or CDS for the window	High (Kadamba), low (Bakul)
LP7 A cost trigger for attention, carried on the oversight line	8; rule at 5	Escalation pathway; facility team	Medium, after LP1
LP8 Operator voice in the menu	6; slot at 5	Menu committee (unconfirmed)	Medium
LP9 A named owner of booked, cooked and eaten (held)	5	CFS and CDS	Low
LP10 What a booking costs: billing and payment basis (held)	5	Billing owner; contract holder	Low
LP11 Surface and test the governing mental models	2 (mindset)	Each belief's holder	Medium
LP12 Change the goal the system actually seeks	3	CFS and CDS	Low to medium

LP1: Build the missing routes. Of Kadamba's 32,162 breakfast registrations in April 2026, 37.1% were eaten.

Yuktāhār's September breakfast ran at 17.7 to 40.2% against 56 to 89% for lunch (confirmed). No route from any record to any rule owner exists, and Yuktāhār's own books disagree on 3 September, 77 eaten against 110. The fix is a standing breakfast view, the ledger, of booked, cooked, eaten and left over per hall, weekday and line, sent to the rule owner and back to the operators under one agreed definition of ”ate”. It also carries a weekly note of exams, events and holidays into the day-before planning meeting, which at Yuktāhār now arrives by word of mouth. It has to be blame-free and accept paper, because Yuktāhār has already dropped a commercial ERP system once (single-sourced). Delivery waits on a receiver, since two different people are on record as head of CDS. A bill line showing breakfasts paid for and not eaten is a student-facing variant, but without a way to respond it would only confirm that the charge cannot be avoided.

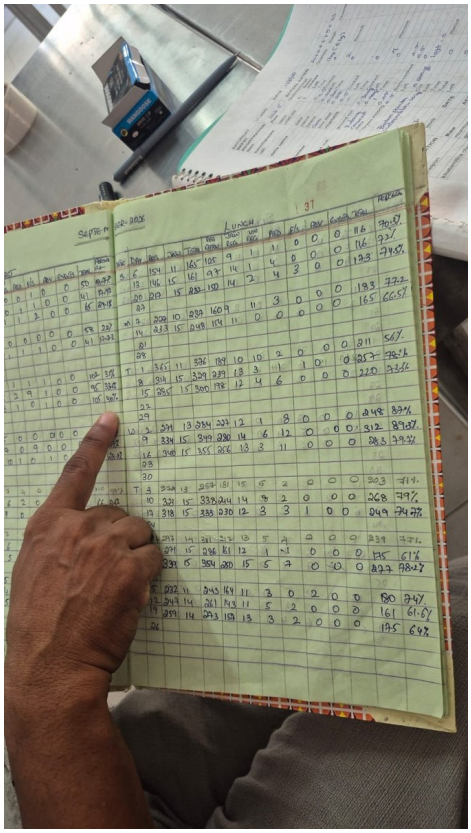


Figure 2. *The records already exist on paper: Yuktāhār's attendance register for September 2026, breakfast on the left page and lunch on the right.*

LP2: Change the breakfast default and its exit. Registrations hold at about 1,070 every weekday while turnout runs from 42.1% on Tuesdays to 27.3% on Sundays (confirmed). The four-day lead protects only dry goods, since Kadamba buys perishables the day before and Yuktāhār takes milk daily, and the Kadamba team has already pushed back on the cut-off (single-sourced). A student and operator coalition for a shorter lead is therefore the first shared interest on record (candidate). The lever is the default and its exit, not the cap. The options, lighter to heavier, are a one- or two-day lead, an evening-before confirm-or-skip that lowers the count, Sunday opt-in, and no registration while off campus. A lower count shrinks Kadamba's first batch only if it is the count the kitchen reads, and whether Skip Meal already lowers it is an open conflict. A skip must never cost a hall slot: one Kadamba respondent books the whole month because the hall fills up (L3b, candidate). The payment basis is unconfirmed for every operator. The Bakul manager says plates served and Kadamba's remark points to registrations (both single-sourced), so Bakul's position on LP2 is likely neutral or favourable if its account holds (candidate). If resale falls, it falls because the need has gone. No absorber is removed.

LP3: Operator goal: surplus is a cost. The purpose would move from ”cook for every booking” to ”feed whoever comes, at least cost” (place 3), with a numeric target inside L11 (place 8). Under the same rules the operators state different purposes. Yuktāhār names quality, less waste, lower cost, standardisation and profit. Kadamba cooks

to registrations ”because it is a responsibility and they are getting paid for it”. The Bakul manager's rule is ”there should not be a shortage of food, even if some wastage occurs” (all single-sourced), and at Bakul a run-out sends a complaint to the managing director while surplus stays silent, a kitchen-level case of a system that reacts only to what is loud (candidate). The fix is a wastage board, a target and a purpose stated by the operations manager, with the target anchored to the best recorded performance (Yuktāhār's rate or the institution's own goal), never last month's average, which drifts down. Smaller portions are the cheapest cut, so every target carries grams per plate, taste and run-outs (P7). A cost-per-line case can go to Prism's operations manager, who has not been interviewed.

LP4: A standing operator forum that runs its own trials. The operators already plan differently, and what separates them is practice more than rules: Yuktāhār's matching-week lookup (books photographed, confirmed), Kadamba's fixed ratio, Bakul's weekly judgement above a floor (both single-sourced). Yuktāhār's waste fell between August and December once weighing was made blame-free (single-sourced, with no before-and-after figures). Kadamba keeps gram standards but no learning record, and the Bakul operator believes Kadamba fills 90 to 95% where the records say 37% at breakfast (confirmed). The fix is a forum convened by CDS and hosted by Yuktāhār's two project managers, where each hall runs its own trial and the ledger shows the results. Swapping books alone is place 6, while a forum with the power to use the ledger to choose what works is place 4. What has to transfer is the preconditions: blame-free weighing, daily discussion and a purpose that treats waste as cost. Kadamba's layered chain adds a reprimand risk.

LP5: Record-calibrated first batch with a daily learning record. L11 closes at Yuktāhār, is a candidate instance bounded by a floor at Bakul, and is absent at Kadamba. Creating it at Kadamba is place 6 and strengthening it where it runs is place 8. Kadamba's 70% becomes 100% for biryani lunch (single-sourced), so the ratio follows expectation and no record. April turnout was 32.4% on the veg line and 48.5% on non-veg (confirmed), so the batch is about 2.2 times what was eaten on veg and about 1.4 times on non-veg (candidate arithmetic). The fix sets the batch as the gram standard times expected turnout per hall, weekday and line, plus a named allowance for outside diners and staff and a float for bonda, puri, vada and idli. It starts on the veg line with a one-page paper record read the next day. A 70% batch against about 398 eaten implies roughly 350 surplus portions (candidate), at odds with Kadamba's stated 5 to 10% waste, so the lever cannot yet be sized. Bakul's manager puts turnout at most around 60%, under his floor of about 70% (single-sourced), and the floor stays a named constraint until item-level records test it. LP5 is a quick fix, paired with LP1 against the L17 risk and with LP6 against run-outs.

LP6: Crest-aware top-up and a staged late batch. At Kadamba 26.5% of April scans land between 9:15 and 9:30, rising to about nine a minute by 9:29, and 9.2%, about 37 a day, arrive at or after 9:30, the latest at 10:07 (confirmed). The operator's estimate of about 150 diners in the last ten minutes overstates this and may drive the hedge (candidate). Yuktāhār projects from the first half hour, when only 15.3% of Kadamba's scans have arrived. The fix has three parts: a weekday arrival card telling the supervisor at 8:00 what share of the day has come (place 6); a late batch started just after 9:00 on that basis (place 10, because it changes when stock is created); and a grace period after 9:30, always paired with the late batch. The share seen by 8:00 runs from 13.5% on Sundays to 22.5% on Wednesdays, and the share after 9:15 from about 32% (Tuesdays, Wednesdays) to about 46% (Sundays), so one flat factor would misread Sunday and Wednesday by about a quarter (confirmed; used as factors, candidate). Nothing here transfers to Bakul, where the manager puts his rush at 8:00 to 8:30 and aims to have food for about 50 at 9:20 (single-sourced). A Bakul top-up needs a phone call, cooking at Hafizpet and a drive of about 8 km, roughly 20 minutes if called early and 40 otherwise, and the end-to-end lead time has never been measured. There the lever is the size of the held reserve, a hedge and not an absorber (candidate), and the latest call that can still land, both set from measured data. LP6 is part of the quick fix.

LP7: A cost trigger for attention, carried on the oversight line. Menu fatigue was fixed through the Student Council, the Parliament and the CDS Committee (confirmed, L4), but the far larger breakfast gap never entered that pathway. The facility team tastes every Kadamba meal with two Prism staff (confirmed, observed) and never checks quantity. The fix is a standing threshold, such as cooked-against-eaten above a set level for several weeks, that sends the case to the Committee as complaints already do. It is anchored to the stated goal, lunch turnout or Yuktāhār's rate, never the hall's own past, and it reports run-outs alongside surplus. The cheapest carrier is one extra line at the tasting. Because an inspector recording quantity can look like an audit, a breach triggers a review and never a deduction.

LP8: Operator voice in the menu. Kadamba's uttapam, set dosa and upma waste most while puri, bonda, vada and

idli move well (single-sourced), yet students or the committee choose the menu, not the kitchen. The Bakul manager wants menu input, knows what sells (pav bhaji) and which pairings he thinks work (kachori with chole), but his view that students would eat more of familiar pairings is untested with residents (single-sourced). The fix is a standing operator slot at each two-week menu cycle to report item waste and run-outs. This builds the missing route L16, the quiet counterpart of L4. Operator input is read with consumption records and checked against residents' views: a voice, never authority.

LP9: A named owner of booked, cooked and eaten (held). Decision rights are split at least five ways, across registration, billing, menu, cooked quantity and the academic calendar (confirmed), and whoever sets registration and billing bears none of the cost they produce. A named owner, not a committee, would put the consequence where the decision is made and inherit LP1. Yuktāhār's Wastage Register keeper is a lighter version. The lever is held until the head of CDS is identified.

LP10: What a booking costs: billing and payment basis (held). Attendance-tracked billing ran during a gas shortage and some holiday and fest periods, and it worked (confirmed). Operators are paid monthly as a lump sum on plates, with deductions (single-sourced, noisy audio), but no contract has been seen and the basis may differ by operator. The lever is held because a billing change made blind to the payment terms could push the operator incentive the wrong way, and student reliance on resale income needs checking first.

LP11: Surface and test the governing mental models. The beliefs holding the default, the fixed ratio and the unrouted gap in place sit at the base of the Iceberg chains, and nobody can change them for someone else. Eight meet a record that does not fit. The four Bakul beliefs meet none yet, so a test record is named for each.

Belief (holder, confidence)	What the record shows	Sign of a shift
The registration count stands in for demand (administration; candidate)	Flat count against 27.3 to 42.1% turnout by weekday (confirmed)	Rule owner cites booked against eaten
”Attendance billing is for emergencies” (candidate)	It ran and worked in the shortage and some fest periods	Discussed for Sunday breakfast
”Cooking to registrations is what we are paid for” (Kadamba; single-sourced)	Yuktāhār plans the same booking from records	Batch set in expected eaters
”Breakfast is about 50%” (Yuktāhār site head; confirmed)	His register shows about 28%	Stated and planned figures match the register
”Student fluctuation is very high” (Yuktāhār; confirmed)	The weekday arrival profile is regular (projection candidate)	A staged cook is tried and tracked
About 150 in the last ten minutes (Kadamba; single-sourced)	About 79 scans a day from 9:20 to 9:29 (confirmed)	The supervisor's prior guess comes close
”We don't have any opportunity to nullify it” (students; confirmed)	The exit exists but is narrow	Skip use rises after LP2
”They are catching us” (Yuktāhār staff; single-sourced)	Dissolved once weighing was made blame-free	Sheets filled unprompted
At most about 60% come (Bakul manager; single-sourced)	No Bakul record. Test: Bakul scans	Batch follows Bakul's scans
Students skip for taste, menu or sleep (Bakul manager; single-sourced)	No resident asked; six students elsewhere cite late sleep most. Test: residents' reasons	Menu changes follow residents' reasons
Pairings, not dishes, are the problem (Bakul manager; single-sourced)	No item or pairing record. Test: cooked and eaten by pairing	A proposal arrives with figures
”There should not be a shortage, even if some wastage occurs” (Bakul manager; single-sourced)	No record of how often a smaller batch would run short. Test: a daily run-out record	Floor and reserve set from that record

Each session starts from the ledger, settles disagreements by the record, pairs advocacy with inquiry and points at specific anomalies such as 50 against 28. At Bakul, which has no record, one has to be built first. Effort goes to the actors already open to the data, and holders of tested views are made visible.

LP12: Change the goal the system actually seeks. The CDS poster gives the institution's goal as minimising food wastage through advance meal planning (confirmed). The goal it actually pursues is the registration count: students are billed on it and Kadamba cooks on it (candidate reading). The stated goal's measure is recorded and read by no one. The fix makes booked-against-eaten and cooked-against-eaten, per hall and line, the real indicators, read

with quality and run-outs, because a wastage goal alone could be met by running short. CFS and CDS carry this by restating their own goal and asking for its indicators. LP12 is the move from the registration count to cooked-against-eaten, and the other levers make that move possible.

Identify System Archetypes

Of six structures checked, only the first three read as present. Shifting the burden orders the whole set.

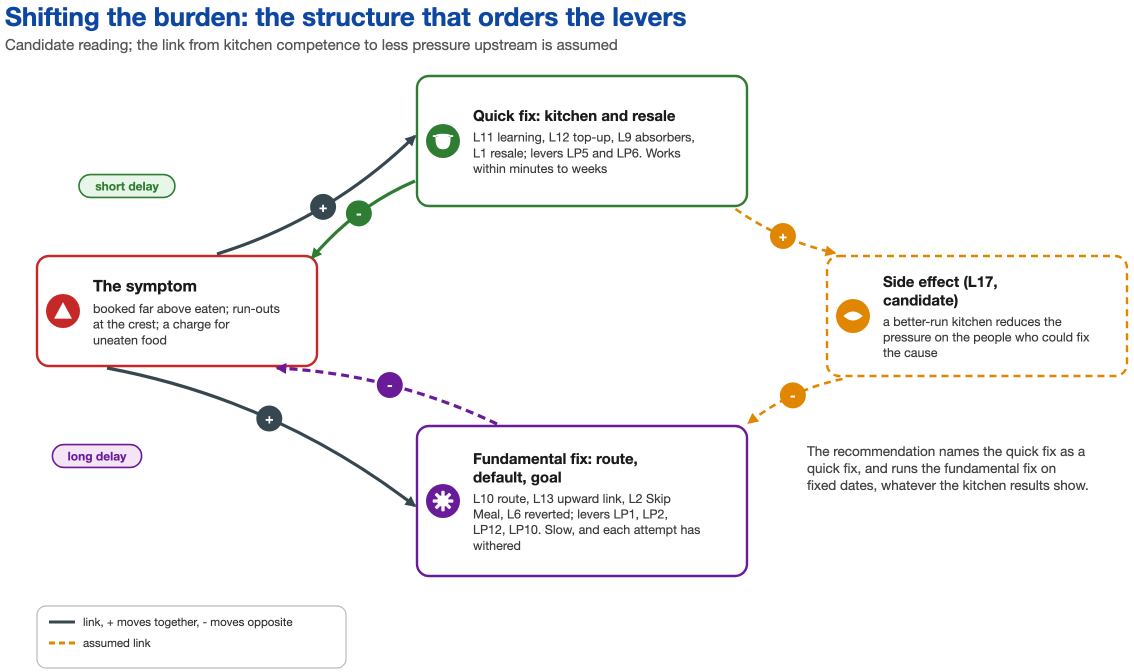


Figure 3. Shifting the burden: the structure that orders the levers.

Archetype	Loops and evidence	Confidence	Way out
Shifting the burden	Quick fix L11, L12, L9, L1; fundamental fix withered (L10 route, L13 upward link, L2, reverted L6); side effect L17. Resale replaces Skip Meal; attendance billing reverted (confirmed)	Candidate; the link from kitchen competence to less pressure upstream is assumed	Strengthen the fundamental fix (LP1, LP2, LP12, LP10), name the quick fix (LP5, LP6)
Balancing process with delay	L12 and L14: a 10 to 15 minute top-up against the crest; at Bakul a 20 to 40 minute drive and a held reserve	L12 confirmed; delays single-sourced; overshoot candidate	Be more responsive without over-correcting (LP6)
Fixes that fail	The reported December 2024 cap tightening; a candidate ratio ratchet on L14; the reheating branch of L13. No backfire observed	Candidate	Make the implicit target explicit (LP3, LP5)
Eroding goals	The stated goal against the count; ”this cannot be helped”; 50 against 28. No lowered target on record	Not established; a risk (assumed)	Anchor standards outside the hall (LP3, LP7, LP12)
Accidental adversaries	Operators and rule owner (L10, L16). No antagonism found	A risk if the ledger reads as an audit	Map the structure together (LP4, LP11)
Commons, or success to the successful	L3b: Kadamba full, Bakul under-registered; no Bakul or Palash records	Low, outside scope	None; LP2's hall-slot condition guards L3b

LP5 and LP6 are the quick fix. They are worth doing because students eat the food saved and they buy time, but the ledger has to label them as such: a better-run kitchen presented as the answer would end the search for the fundamental fix (the L17 risk). LP1, LP2, LP11 and LP12, with LP10 once released, form the fundamental fix, and LP12 answers the sharpest finding of the problem analysis, that the system chases the wrong goal. The one past change that pushed the wrong way is the December 2024 tightening of the cap to ease Mess Office workload, which narrowed the exit (single-sourced; effect on no-shows candidate).

Ranking: Depth and Feasibility

Depth combines place with role, so a fundamental-fix lever outranks a quick fix at the same place. Feasibility is judged for a student team over the weeks remaining.

Rank	Lever (place)	Role	Depth / feasibility	Gated on
1	LP1 (6)	Fundamental, enabling	High / high to prototype	A named receiver
2	LP11 (2)	Fundamental	Very high / medium	The ledger
3	LP2 (5)	Fundamental	High / medium	Lowering the kitchen's count
4	LP12 (3)	Fundamental	Very high / low to medium	CFS and CDS
5	LP3 with LP4 (3, 8; 4)	Links both	High / low to medium	Access to Prism management
6	LP5 with LP6 (6, 8; 6, 10)	Quick fix	Medium / high to low by hall	Waste base; Bakul records
7	LP7 (8)	Fundamental	Medium / medium	Facility team's parent body
8	LP8 (6)	Fundamental	Medium / medium	Records and resident views
Held	LP9 (5)	Fundamental	High / low	Head of CDS
Held	LP10 (5)	Fundamental	High / low	The plate basis

LP1 ranks first, though shallower than LP11, LP12 and LP2, because every other lever needs its input. LP11 is the deepest but needs the ledger first. LP2 is the deepest rule lever not gated on the plate basis. LP12 sits fourth because nobody who could carry it has engaged, and a goal without indicators or an exit does nothing alone. LP3 and LP4 carry the goal into the operator's routine, since Kadamba has the information but no reason or routine to use it. LP5 and LP6 are the fastest physical levers and land at Kadamba only through LP3 and LP4. LP7 outranks LP8 because it gives governance a response to cost. LP9 and LP10 are held, not dropped.

1.2 Explore Changes to Rules, Roles, Incentives, Information Flows and Processes

Type	Today	Candidate change	Lever
Rules	Automatic registration; five cancellations a month, four days ahead; back-door entry after 9:30 cannot be refused (single-sourced); billing at the booking	Shorter lead; evening skip that lowers the count; Sunday opt-in; no hall-slot penalty; consumption billing as already run; a grace period paired with a late batch	LP2, LP10, LP6
Roles	The facility team checks safety and taste only; nobody convenes operators or owns the outcome; the menu is chosen without operator input	A quantity line at the tasting; a CDS-convened forum; a named owner; an operator menu slot; CFS and CDS as carriers of the goal	LP7, LP4, LP9, LP8, LP12
Incentives	A non-eater pays anyway; Skip Meal returns nothing; recording is read as being caught; shortage is loud and surplus silent (candidate)	A later exit that removes the charge; surplus shown as cost; blame-free recording; a late batch or a sized reserve	LP2, LP3, LP5, LP6
Information flows	Records reach no rule owner; two definitions of ”ate”; the calendar by word of mouth; no kitchen-to-menu route (L16)	The ledger; a weekly calendar note; item waste into the menu cycle	LP1, LP8
Processes	Fixed 70% at Kadamba, a weekly judgement with a floor at Bakul, no record; the whole hedge cooked at the start	A calibrated batch with a daily record; an arrival card; a late batch after 9:00; safety steps unchanged	LP5, LP6

The widest rule change costs the kitchen least, because the four-day lead only protects dry goods. The incentive finding depends on how plates are paid. On registered plates, cutting surplus lowers cost while revenue holds: a route open now that closes if the rate is renegotiated, and turns harmful if surplus is cut through smaller portions. On served plates, the gain and the pressure to serve more both grow. If the Bakul manager's account holds, Bakul already absorbs the cost of uneaten portions to avoid shortage (candidate). P7's paired measures cover both cases.

The Natural Comparison

Kadamba and Yuktāhār share registration, billing and, as far as is known, tendered plate payment, yet Yuktāhār plans from records, re-cooks by run-out time and weighs blame-free on paper, while Kadamba cooks a fixed ratio. A learning practice is therefore possible under current rules.

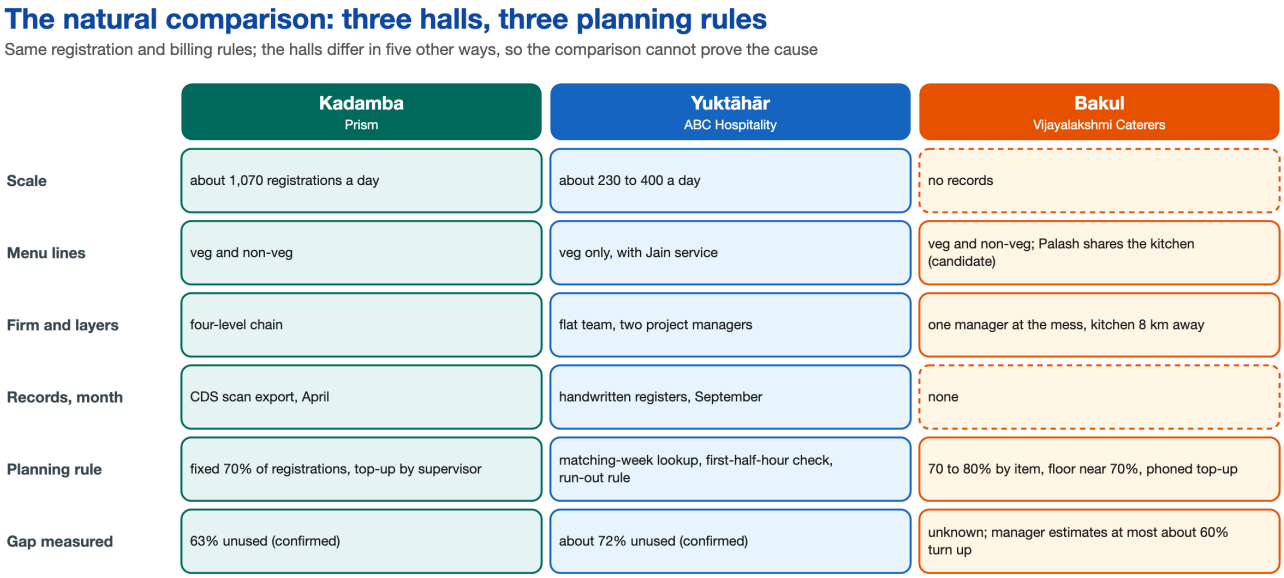


Figure 4. *The three halls side by side.*

The halls differ in five other ways, so the comparison cannot prove the cause. Two claims stay candidate: that the difference is goal and culture, and that LP3 through LP6 work without a rule change. The gap itself is similar at both recorded halls, so kitchen practice reduces waste and run-outs but only LP2, LP10 and the goal and belief levers move the booking. Bakul, a third regime with its own rival explanations (an off-site kitchen and a different payment report), needs its own records before it joins the comparison.

1.3 Develop Design Principles

The objective is a breakfast system where registration matches each student's intent, the kitchen cooks to expected eaters, and the cost of any gap reaches whoever sets the rules. It does not mean ending registration, resale or all waste. Registration gives the planning count, resale recovers money, and some surplus is the price of never running out. Over-booking and over-cooking should become noticed exceptions. The guiding idea from the archetypes is to change the architecture instead of competing at the kitchen layer.

The status quo persists because it pays each actor. The levers keep what each actor values (the seat, the resale, the absorbers, run-out protection) and change what drives over-booking.

Actor	What the status quo gives	Cost of change	Cost of not changing
Students	No decision; a guaranteed seat; resale income	A nightly decision; less resale	Paying for uneaten breakfasts (63% of Kadamba's April registrations); reheated food
Prism (Kadamba)	Revenue possibly on registered plates; few run-outs	Possible revenue fall; a daily record	Food cooked that no one eats; reheating
Bakul operator	No shortage, secured by a floor and reserve (single-sourced)	A check on the manager's judgement	Food held hot for hours, softening items such as aloo paratha
CDS	Low workload	A standing ledger; portal changes	A stated goal it cannot show it meets
Staff and outside diners	Food from surplus	Meals must be budgeted	Meals stay a by-product
CFS	A predictable budget	A budget that varies with consumption	Spending on food nobody eats

CDS and the Mess Office gain most from low workload and own the rules that the fundamental fix touches, so LP1, LP11 and LP12 have to reach them before any rule proposal.

1.4 Identify Potential Intervention Points

The fourteen intervention points follow one breakfast from booking to bin, then its governance. IP1 is the default, IP2 the cancellation exit and IP3 the count the kitchen actually reads.

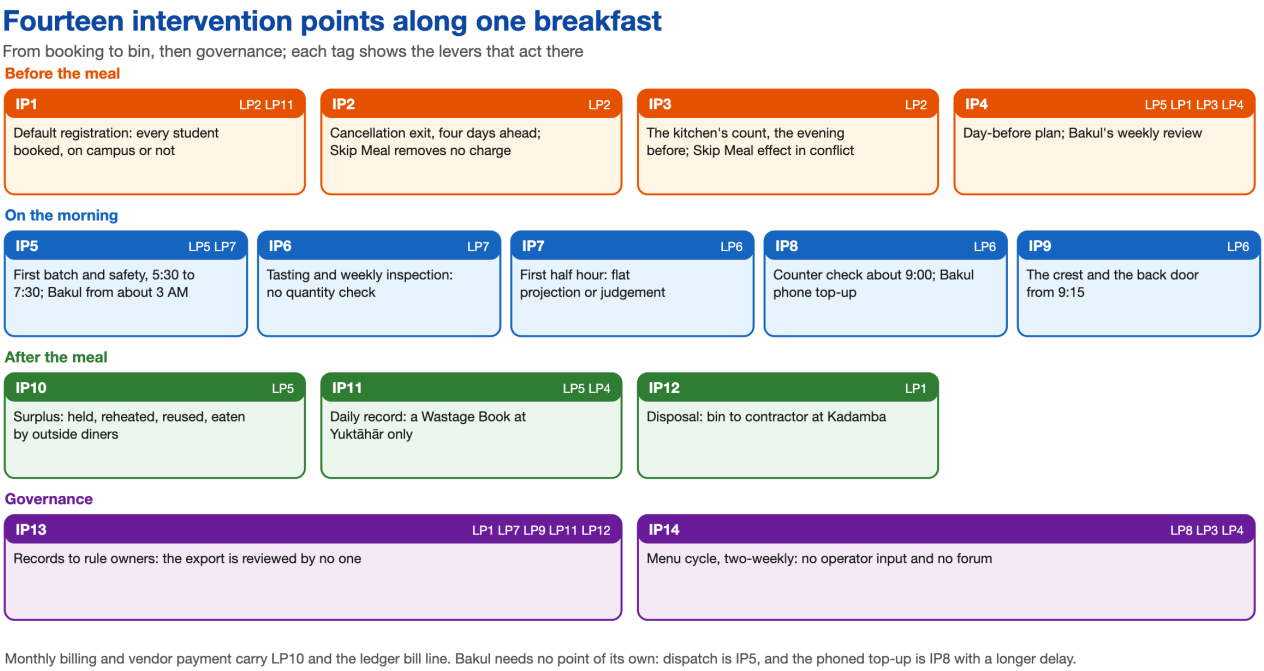


Figure 5. Fourteen intervention points, with the levers that act at each.

Sunday Breakfast: Where Three Levers Meet

Sunday has Kadamba's lowest turnout. A 70% veg batch is about 3.1 times what Sunday diners ate, against about 1.9 times on Tuesdays (candidate arithmetic), and Yuktāhār's Sundays run at 17.9 to 24.1% (confirmed). LP2 (IP1 to IP3), LP5 (IP4) and LP6 (IP8, IP9) bite hardest on Sunday, which makes it the natural first slice for Activity 10. The Bakul manager calls weekends ”very low” but gives no figure, so the slice stays specific to Kadamba's veg line.

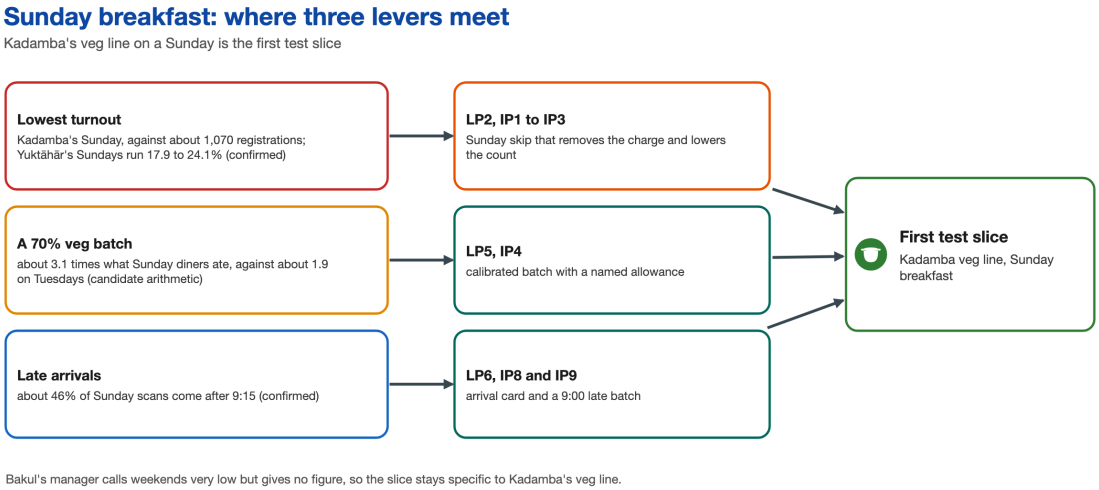


Figure 6. Where the three levers meet on a Sunday.

1.5 Deliverable 1: Leverage-Point / System Intervention Map

The map draws breakfast as one chain from the automatic default through the day-before plan, first batch, service and surplus to the bin, with records and money above and governance on top. Markers are coloured by place. Dashed red lines are routes that do not exist today and that a lever would have to build, and the dashed green learning link runs at Yuktāhār and is absent at Kadamba. Grey tags carry the intervention points.

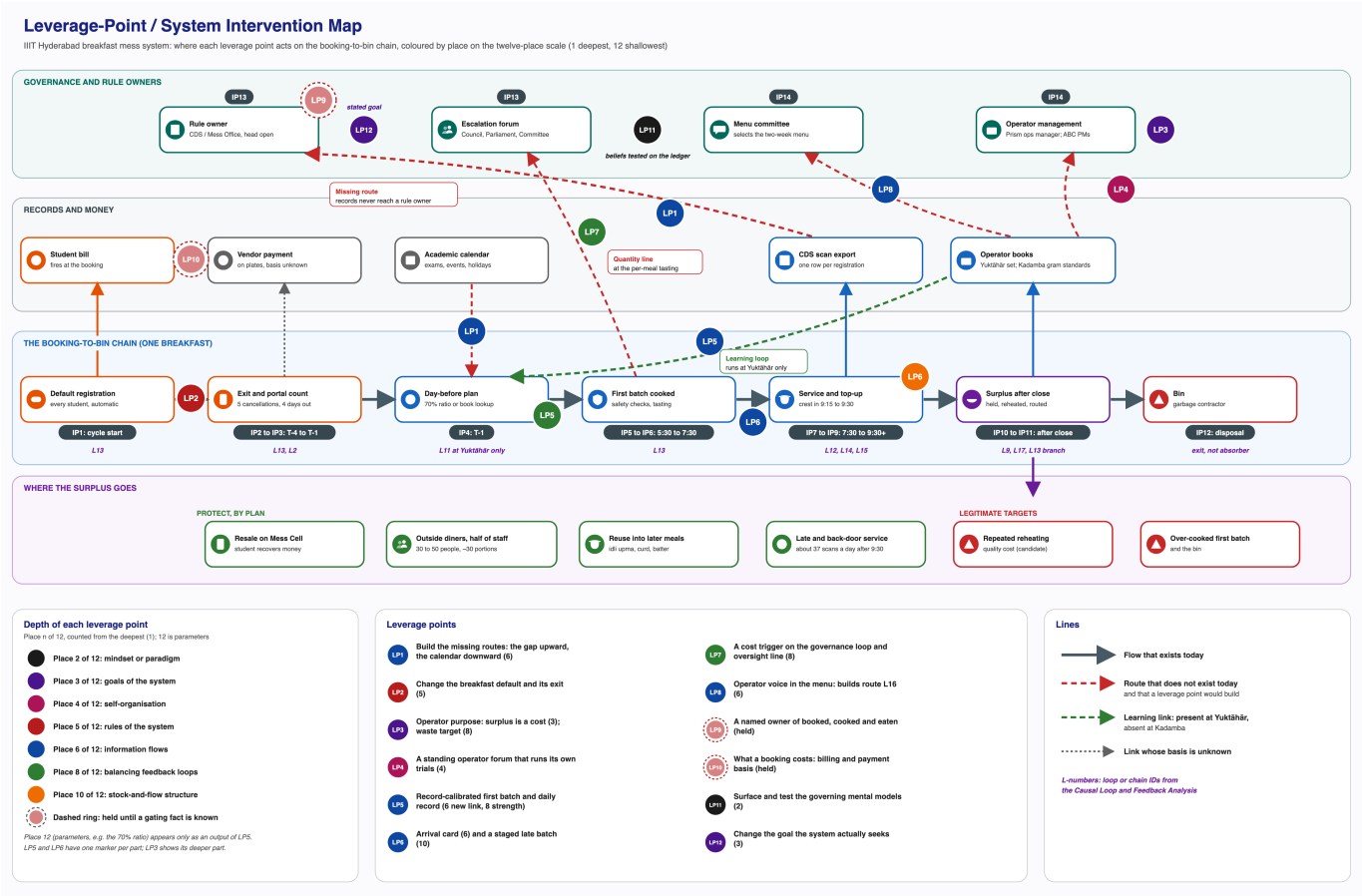


Figure 7. Leverage-point and intervention map for the breakfast mess system.

Every solid flow runs along the chain or up into a record, and stops there. The levers that build a route, set a goal, test a belief or convene operators sit on routes or bodies not yet doing that work, and the rule levers sit on the booking, the bill and the owner those routes would reach. The learning link, confirmed at one hall of three, is the only working feedback into the day-before plan.

Intervention point	Levers (place)	Holder of the power	Absorber check
IP1 to IP3 Default, exit, kitchen count	LP2 (5); LP11 students' belief (2)	Rule owner and portal owner; student representatives	Resale need shrinks; none removed
IP4 Day-before plan	LP5 (6 Kadamba; 8 Yuktāhār, Bakul); LP1 calendar (6); LP3 (3, 8); LP4 (4); LP11 operator beliefs (2)	Operators; academic administration via CDS; Prism management	Budget outside diners, staff and reserve
IP5, IP10, IP11 Batch, surplus, record	LP5 (6, 8); LP4 (4)	Operator	Keep safety steps and raw reserve; diners fed by allocation
IP6 Tasting and inspection	LP7 (8; 5 for the rule)	Facility team; escalation pathway	Adds no delay to safety
IP7 to IP9 Projection, top-up, crest	LP6 (6, 10)	Supervisor; Yuktāhār team; Bakul manager; Mess Office or CDS for the window	Keeps back-door service, made explicit; Bakul reserve sized
IP12, IP13 Disposal, records upward	LP1 (6); LP7 (8); LP9 (5); LP11 (2); LP12 (3)	CDS, receiver unnamed; CFS and CDS	The bin is a target

Intervention point	Levers (place)	Holder of the power	Absorber check
IP14 Menu and management	LP8 (6); LP3 (3, 8); LP4 (4); LP11 (2)	Menu committee (unconfirmed); Prism management	Removes nothing
Monthly billing	LP10 held (5); LP1 bill line (6)	Billing owner; contract holder	Check reliance on resale

Only LP5 and LP6 touch an absorber that protects someone, and both are written to keep it intact. LP5 budgets outside diners and half of Kadamba's staff by name and keeps reuse, the walk-in buffer and the top-up reserve. LP6 keeps them too and makes late and back-door service explicit with a late batch. LP2 and LP10 shrink resale by removing the need for it, not the market itself. LP1 records repeated reheating and the bin, LP3 through LP6 cut them directly, and LP2, LP10, LP11 and LP12 act upstream. No lever removes an absorber to create a signal, though that does not mean no lever costs anyone anything.

Stakeholder	Gains	Bears	Levers
Students	A later exit; less reheated food; fewer late run-outs; an honest bill	A nightly decision; possible loss of resale income	LP1, LP2, LP5, LP6, LP10, LP11
Prism (Kadamba)	Lower food cost; fewer late shortages	Possible revenue loss and rate renegotiation; a daily record	LP2, LP3, LP5, LP6, LP10
Kadamba supervisor, staff, storekeeper	A clearer late-batch rule; issues matched to a plan	A 9:00 cook decision, a daily sheet, blame risk, a varying batch	LP5, LP6, LP7
Yuktāhār operator	A menu voice; a better breakfast anchor	Hosting time; its head's figure set against his own register	LP4, LP5, LP8, LP11
Outside diners and Kadamba staff	Meals by allocation, not accident	Being squeezed if the allowance lapses	LP5
Rule owner (CDS or Mess Office)	The gap visible in its own terms	A ledger, a threshold to answer, portal changes, an ownership role	LP1, LP2, LP7, LP9, LP12
CFS and facility team	Evidence the goal is met; a remit reaching quantity	A less predictable budget; being seen as auditors	LP7, LP10, LP12
Bakul operator	A check on its judgement; a menu voice; no revenue loss from LP2 if paid on plates served (candidate)	Requests for records it may not keep; any rise in shortage risk	LP2, LP3, LP5, LP6, LP8, LP11

Each actor has a rational reason to keep the present arrangement, so resistance is expected and is not a fault to fix. Kadamba's operator loses most from LP2 yet is most needed for LP5, so at Kadamba LP3 and LP4 come before LP5. At Bakul the likeliest objection is to anything that raises shortage risk (candidate, held behind the unconfirmed plate basis).

1.6 Deliverable 2: Design Principles

Seven principles follow from the leverage points, archetypes and evidence, and every Activity 10 concept must pass all seven.

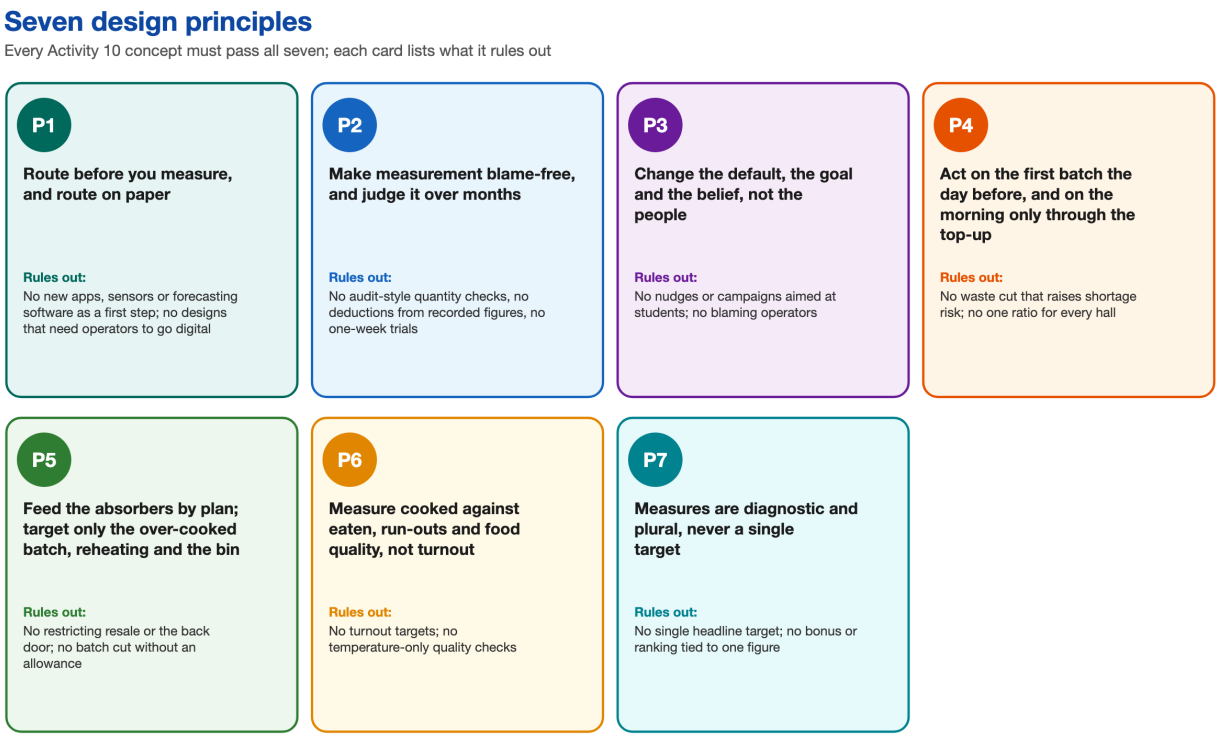


Figure 8. The seven principles, with what each rules out.

P1: Route before you measure, and route on paper. The scan export, Yuktāhār's books and Kadamba's own stated waste share already measure the gap, so connect them to the decisions that need them before adding any instrument. Yuktāhār dropped a commercial ERP over cost and staff readiness (single-sourced), so routes start from a book page and use existing channels: the daily meeting, the tasting, the menu cycle.

P2: Make measurement blame-free, and judge it over months. No recorded figure may justify a deduction against the staff who recorded it. Yuktāhār's weighing worked only once it was blame-free and discussed daily, and waste took from August to December to fall (single-sourced). One run-out after a smaller batch, judged over a week, would bring the hedge straight back.

P3: Change the default, the goal and the belief, not the people. Treat every actor as rational given its incentives and information. Students are registered by a default they never chose, operators run the only working balancing loops (L11, L12), and the institution states one goal and pursues another. Nudges and campaigns aimed at students, blaming operators, and changing the cancellation count while the default stays untouched are ruled out.

P4: Act on the first batch the day before, and on the morning only through the top-up. Morning levers work through smaller, later top-ups or, where a top-up cannot land in time, a sized reserve. No lever may cut waste by raising shortage risk, and scope stays per hall, weekday, line and, where records allow, item. Kadamba fixes the batch three and a half to four hours before the crest, Bakul can start cooking around 3 AM (single-sourced), and arrival curves and line turnouts differ widely. One hall's rule is not carried to another before that hall has records.

P5: Feed the absorbers by plan, name them as symptomatic outlets, and target only the over-cooked batch, repeated reheating and the bin. A concept that shrinks surplus must name who eats it and budget their meals. Thirty to fifty outside diners and about half of Kadamba's staff already eat this food (single-sourced), and calling their meals ”not wasted” would hide the gap again, the shifting-the-burden trap in miniature. Restricting resale, closing the back door and cutting the batch without an allowance are ruled out.

P6: Measure success as cooked against eaten, run-outs and food quality, not turnout. Success is the cooked-eaten gap, run-outs, food quality as eaten (holding time and texture as well as temperature and taste) and whether the cost reaches a rule owner. Falling waste with rising run-outs does not count. Some skipping is legitimate

(confirmed across interviews), and Bakul's food, held at about 60 to 72 °C, keeps its temperature while aloo paratha softens (single-sourced). No turnout targets, no waste cuts sized before Kadamba's 5 to 10% base is known, and no temperature-only quality checks.

P7: Measures are diagnostic and plural, never a single target. Measures are for learning and never for reward or penalty. Cooked-against-eaten always comes with grams per plate, taste and run-outs, because a kitchen held to one figure can meet it by cutting portions or letting the crest run short, and a vendor rewarded on it can meet it through the rate. The targets in LP3 and LP7 serve review and are not to be gamed. A single headline target, any bonus, deduction or ranking tied to one ledger figure, and waste figures reported alone are ruled out.

Phase 4: Design, Test and Evaluate Interventions (Activity 10)

The Breakfast Paradox: IIIT Hyderabad Breakfast Mess System, Team Invictus

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2 Activity 10: Design, Test and Evaluate Interventions

Activity 10 tests interventions against the whole system, not one actor's interest. The concepts aim at the three confirmed structures that hold the paradox in place (the automatic default with a narrow exit, billing at the booking, and a gap that is measured and never routed) and at the step where registrations become a first batch. Every scenario output is labelled *projected*: arithmetic on real records under stated assumptions, never an observed result. Stakeholder responses are *simulated* and co-design tests are *planned*; none has been run. All modelled figures are Kadamba's, because Bakul has no records.

2.1 Generate Alternative Interventions (Deliverable 3: Intervention Concepts)

Eight concepts were generated. Seven aim at a lever; G is a deliberate foil, kept in every test so that its failure shows in the numbers.

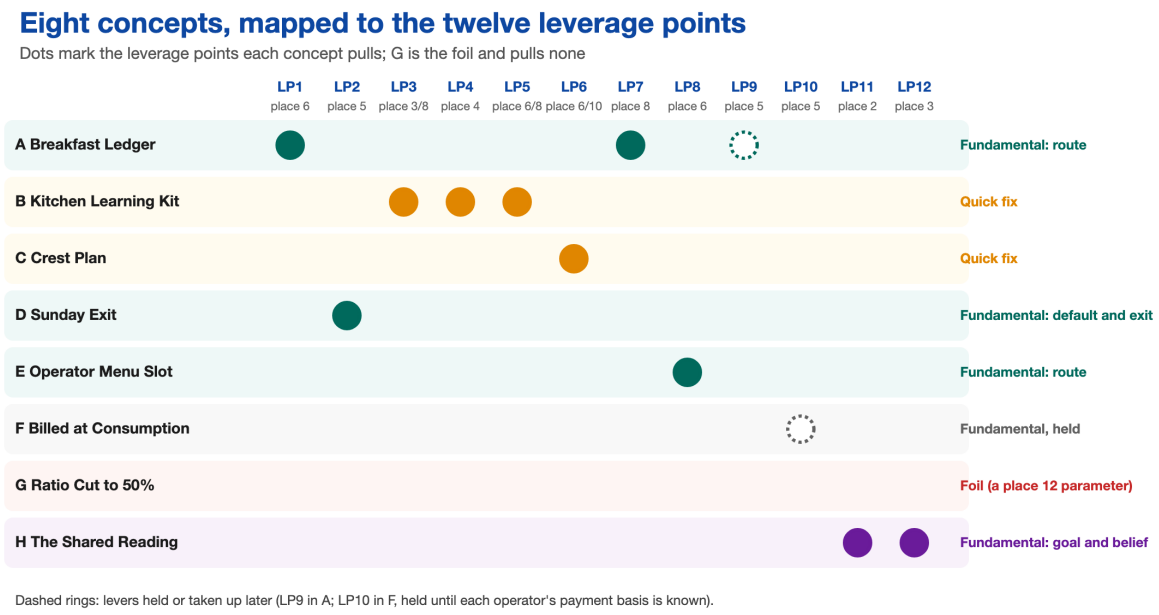


Figure 1. Eight concepts mapped to the twelve leverage points.

A: The Breakfast Ledger. Each week the scan export and the operators' books merge into one sheet per hall, by weekday and line, with the money charged for uneaten breakfasts. Kitchen levers shrink surplus but leave that charge flat until a rule moves, so a falling surplus cannot pass for ”problem solved”. A cooked-against-eaten threshold, anchored outside the hall's own past, alerts the escalation forum next to the existing complaint trigger, and a weekly note of exams and events goes to planning. A builds the missing route (L10), adds a second trigger to L4, counters L17 and brings L7's exam dips to the kitchen (P1, P2, P6, P7).

B: The Kitchen Learning Kit. Kadamba's fixed ratio becomes a day-before plan card built from expected eaters, with the 30 to 50 outside diners and about half of Kadamba's staff allowed for by name, and a daily paper sheet read the next day. An operator forum, convened by CDS and hosted by Yuktāhār's project managers, carries over what makes Yuktāhār's Wastage Book work: blame-free weighing, daily discussion and surplus treated as cost. At Bakul, whose manager already adjusts weekly by item, the kit works as a check more than a rule, and the floor stays until a record shows how often a smaller batch would run short. B creates L11 at Kadamba, shortens L13's kitchen half and reaches L8 through cost (P1 to P7).

C: The Crest Plan. At 9:00 the supervisor projects the day from a weekday arrival card and starts a late batch that lands by about 9:15, so the first batch covers only the earlier diners. A grace period after 9:30 covers only late-batch food. All of it depends on Kadamba's 10 to 15 minute on-site top-up. A Bakul top-up needs a phone call, cooking at Hafizpet and a 20 to 40 minute drive (single-sourced), so its hedge is the roughly 50 portions held at 9:20 (candidate), and C there means sizing that reserve and the latest call that can still land. C gives L12 sight of the crest, removes L14's reason to hedge and bounds L15 (P4, P5, P6).

D: The Sunday Exit. Sunday has the lowest turnout at both recorded halls. It is drafted first as opt-in, charged only if confirmed by 8 pm on Saturday, with a student who forgets paying the walk-in rate. A lighter variant keeps the default and adds a skip, open until the same hour, that removes the charge and lowers the kitchen's count. Other days keep the four-day lead and the cap of five, and no skip may cost a hall place. D acts on the booking end of L13, on L2 (Skip Meal fails today because a skip returns nothing) and on L1 (P3, P4, P5).

E: The Operator Menu Slot. At each fortnightly menu cycle operators bring the items that wasted most (uttapam, set dosa and upma at Kadamba, single-sourced) and those that ran out. The Bakul manager thinks pairings are the real problem (single-sourced), so operator input counts as a voice and never authority, read beside consumption data and residents' wishes. E supplies L16's missing link (P1, P6).

F: Billed at Consumption (held). Breakfast is charged on the scan, as it was during an LPG shortage and some fest periods (confirmed), acting on L1, L2, L6 and L8 (P3, P5, subject to a check on resale income). It is held until each operator's payment basis is confirmed, since Bakul's manager reports plates served while Kadamba's operator says it cooks on, and is paid for, registrations (both single-sourced).

G: Ratio Cut to 50 Percent (foil). Kadamba's first batch drops from 70 to 50% by instruction, with no record, late batch or allowance. It changes no loop and breaks P4 and P5.

H: The Shared Reading. Joint sessions around the ledger ask each belief holder what produces their figure and what would change it: the Yuktāhār site head's ”about 50 percent” against his own register at about 28, the Kadamba supervisor's crest prediction, and the registration side's reasons for the automatic default. CFS and CDS are asked to adopt the CDS poster's existing goal, minimising food wastage through advance planning, as two indicators, booked-against-eaten and cooked-against-eaten, read beside P7's quality measures. The Bakul manager's beliefs wait on a Bakul record. H acts on the mental models under every chain and on the working goal behind L10, L13 and L17 (P2, P3, P6, P7).

2.2 Develop Scenarios and Prototypes (Deliverables 3 and 4: Prototypes and Scenario Model)

The prototype is the re-sequenced bundle across one breakfast. The kitchen lane is the quick fix. The student, records and governance lanes carry the fundamental fix on fixed dates.

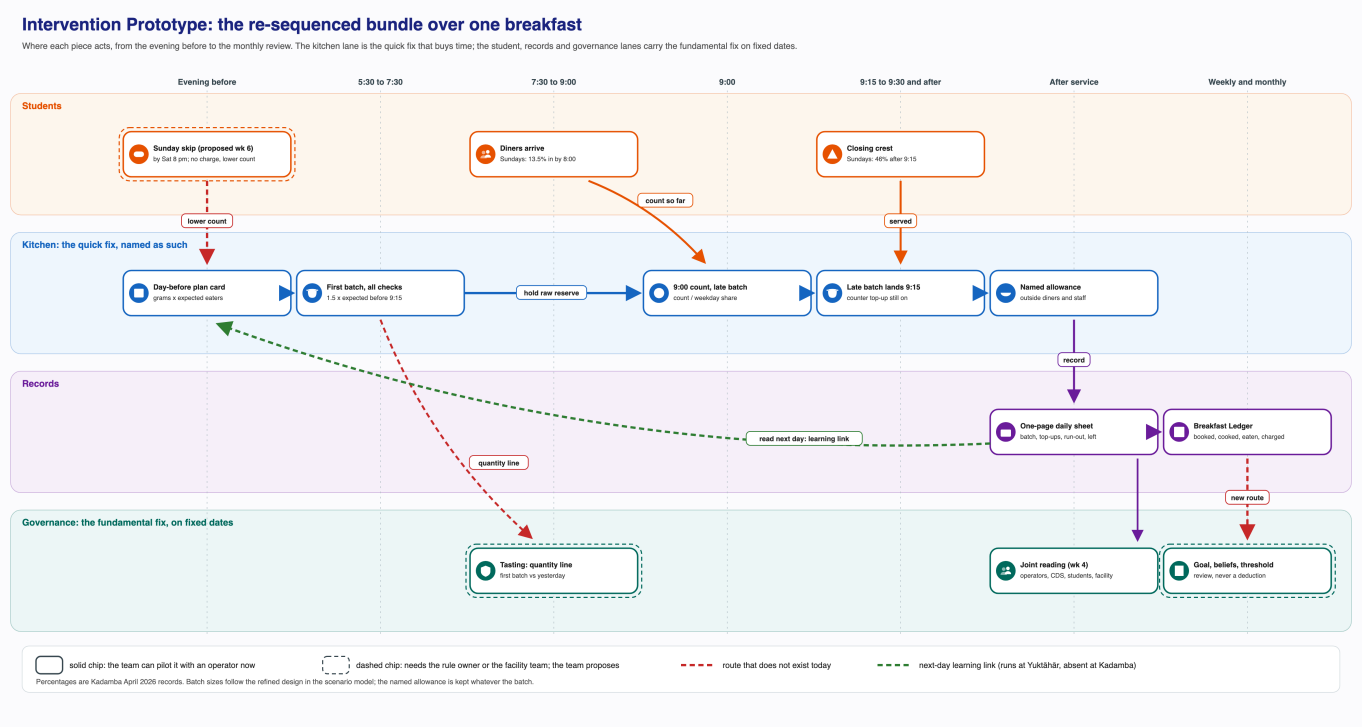


Figure 2. The re-sequenced bundle across one breakfast: quick fix and fundamental fix on separate lanes.

Prototypes

Prototype A: the ledger sheet. One A4 page per hall per week, filled from the scan export and the operator's daily sheets, handwritten if need be and typed later, as Yuktāhār does with its books (single-sourced). Filled for Kadamba on Sunday 19 April 2026: veg line, 817 registered, 179 eaten, 638 charged and not eaten at Rs. 48, which is Rs. 30,624; non-veg line, 252 registered, 94 eaten, 158 at Rs. 66, which is Rs. 10,428. Registered and eaten are confirmed; the charge applies posted rates to April and is projected. Four rules sit at the foot of the page. ”Ate” means a scan against a registration, with hand-logged scanner failures shown separately. On a run-out day eaten is a floor. The uneaten charge is printed beside the surplus. No figure on the page may ground a deduction.

Prototype B: the plan card and daily sheet. The card is one line per dish per line, filled at the evening planning meeting from records available by Saturday evening. Grams follow the existing standard (four idlis a person from 120 grams of batter, 100 grams of sambar; single-sourced). The daily sheet logs each batch and top-up, the run-out time, and the leftovers and where they went. At Bakul the prototype is a per-item sheet only. Yuktāhār already runs its day-before step on paper.

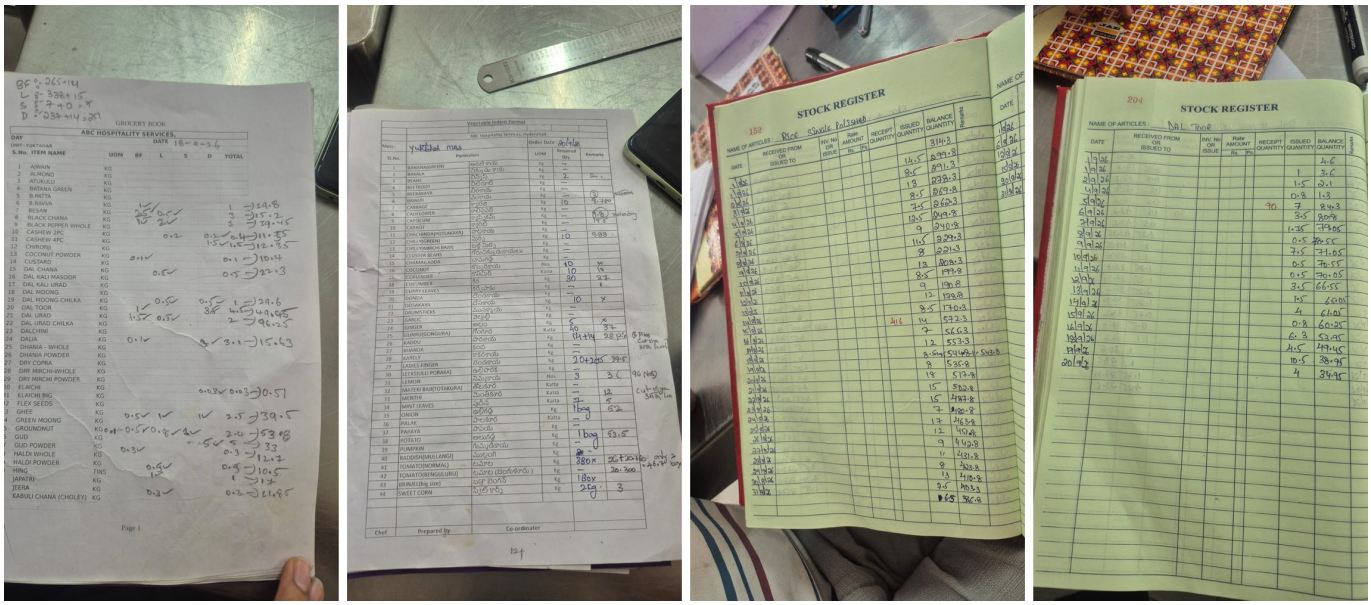


Figure 3. The paper formats Prototype B borrows: Yuktāhār's Grocery Book, vegetable indent and stock registers (confirmed, photographs).

Prototype C: the arrival card and the 9:00 rule. The card is printed from the April records (confirmed). At 9:00 divide the diners so far by the day's usual share by then, add 10%, subtract the first batch and cook the difference. The grace period is withdrawn if scans after 9:30 stay above April's 9.2% for four weeks running.

Weekday	In by 8:00	By 9:00	By 9:15	After 9:15	Scans a day, 9:15 to 9:30
Monday	17.7%	50.7%	64.6%	35.4%	117
Tuesday	17.0%	55.2%	68.5%	31.5%	110
Wednesday	22.5%	54.6%	68.7%	31.3%	93
Thursday	21.9%	52.9%	64.5%	35.5%	108
Friday	13.9%	48.2%	62.7%	37.3%	106
Saturday	15.8%	49.4%	62.2%	37.8%	106
Sunday	13.5%	38.9%	54.4%	45.6%	103

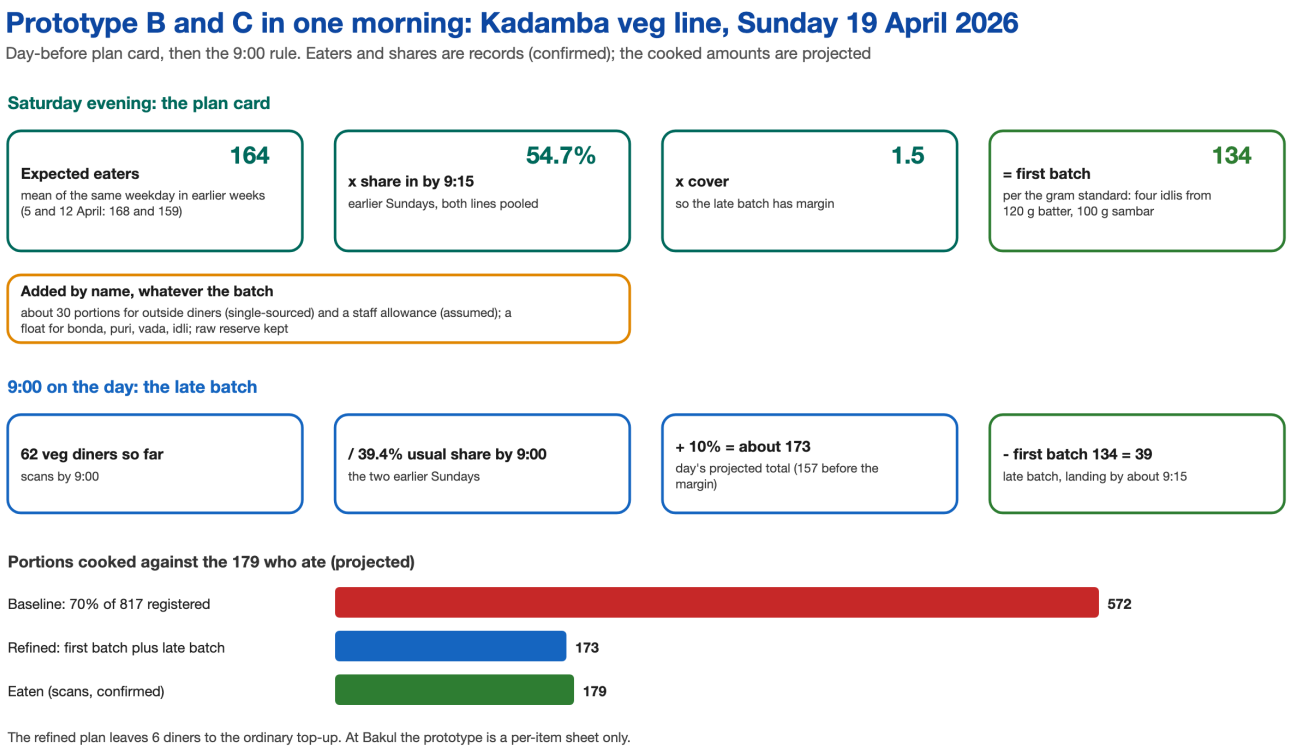


Figure 4. *Prototypes B and C on one morning: the plan card, the 9:00 rule, and portions cooked against portions eaten.*

Prototype D: the Sunday exit on the portal. One control on the existing Skip Meal screen for Sunday breakfast, open until 8 pm on Saturday, in two forms. The first-draft opt-in reads ”Confirm Sunday breakfast: you will be charged only if you confirm” and the kitchen reads the confirmed count. The evening-before skip reads ”Skip Sunday breakfast: you will not be charged, and the kitchen will cook less”, removes the charge, and the kitchen reads registered minus skips. Whether Skip Meal already lowers the kitchen's count is an open conflict, so the prototype requires that either form does.

Prototypes E and H. The menu note is half a page per operator per cycle: the three items most left over, the three that ran out earliest, items above the gram standard, and residents' views beside any proposed pairing. In the hour-long shared reading each participant writes their own figure before the page is read together, and the session records, in their words, what would show the belief has shifted. Nothing said is attributed outside the room.

Co-design and User Testing (Planned)

These tests are planned for Stage 1, before any pilot. They would catch a design nobody would use, which no model can do. The Bakul tests prepare a later extension and do not gate the Kadamba pilot.

Test (planned)	With whom	What it checks	Result that would change the design
Sunday skip wording	8 to 12 students: skippers, resellers, daily eaters	Understood at once; fear for slot or resale; preferred variant	Misreading: rewrite. Slot fear: guarantee on screen. Preference for confirming: opt-in later
Plan card and 9:00 walk-through	Kadamba supervisor and cook	Card fits the evening step; 9:00 batch without delay; 15-minute items	Cannot land by 9:15: 8:45 trigger, higher cover. Too slow: three main veg items only
Ledger review	Yuktāhār's managers and register keeper	Audit or shared record; fit with paper routine; which ”ate”	Audit reading: no per-person entries, monthly threshold. Mismatch: their columns as template
Export effort	A CDS staff member	Time for the export and the page	Over an hour a week: monthly page
Why Bakul residents skip	8 to 12 Bakul residents	Their reasons against the manager's; pairings; held items	Mostly sleep: menu is no attendance lever. Texture complaints: record holding time

Test (planned)	With whom	What it checks	Result that would change the design
Bakul reserve walk-through	Bakul manager	Call-to-counter time; basis of the reserve; whether anything is written	Long lead: reserve kept, sized from arrivals. Nothing written: per-item sheet first

Scenario Model and Impact Evaluation

The model explores how the system might respond and makes no predictions. It replays Kadamba's April 2026 breakfast records (confirmed: 32,162 registration rows with scan times), setting what each rule would have cooked against what was eaten, arrival by arrival. Kitchen forecasts use only earlier records, so the backtest covers 23 out-of-sample days (8 to 30 April); student-side scenarios use the whole month. Money is the posted rate times counted uneaten registrations, and no food cost or waste weight is used. ”Surplus” here means portions cooked beyond what registered students ate, which is not waste in the strict sense. Demand is censored, since on a run-out day ”eaten” is a floor and an unadjusted forecast would drift down, and the ledger's run-out rule corrects for that.

The inputs: a 70% first batch per line (tested 60 to 80, single-sourced); posted rates of Rs. 48 veg and Rs. 66 non-veg at Kadamba and Rs. 53 at Yuktāhār (confirmed as posted, use for April assumed); a forecast equal to the mean of earlier same weekdays; arrival profiles from earlier same weekdays, with the crest 5 points lighter to 10 heavier in stress runs; a 15-minute top-up (30 in a gas shortage); an allowance of about 30 outside diners plus 0 to 30 staff; skip uptake of 25 or 50% of no-shows (tested 10 to 75, assumed); opt-in turnout of 60, 72 or 85%; eaters held at records (±10%, ±25% for exams or a favourite item); and adoption on assumed dates. The records cover one hall, one meal and one month, and rest on the operator's own ratio.

Bakul's account differs on every kitchen input: a rush at 8:00 to 8:30 with 10 to 15 arrivals in the last ten minutes, cooking about 8 km away from around 3 AM for some items, an unmeasured top-up lead time, and turnout at most about 60% (candidate, possibly itself an upper bound). The arrival card, the 9:00 batch, the 1.5 cover and the Scenario 4 thresholds apply to Kadamba only.

Scenario 1. The kitchen rules, backtested. The baseline cooks far ahead of demand, with the veg line at about 525 a day for 237 eaten. The ratio cut fails both ways: 138 of its 150 surplus portions are still veg, and non-veg, whose turnout reaches 59% on busy days, runs short on 7 of 23 days, because one flat ratio cannot serve two lines 16 points apart. A calibrated single batch cuts surplus to 60 but moves the cost to the crest: 16.8 short a day, and on 26 April the veg line would have run out at 9:24, 45 short, L14 in reverse. Adding the late batch cooks 274 fewer portions than the baseline and leaves 91 surplus and 2.6 short, which the ordinary top-up covers. The effective ratio becomes an output, from 29% (Sunday veg) to 67% (Friday non-veg).

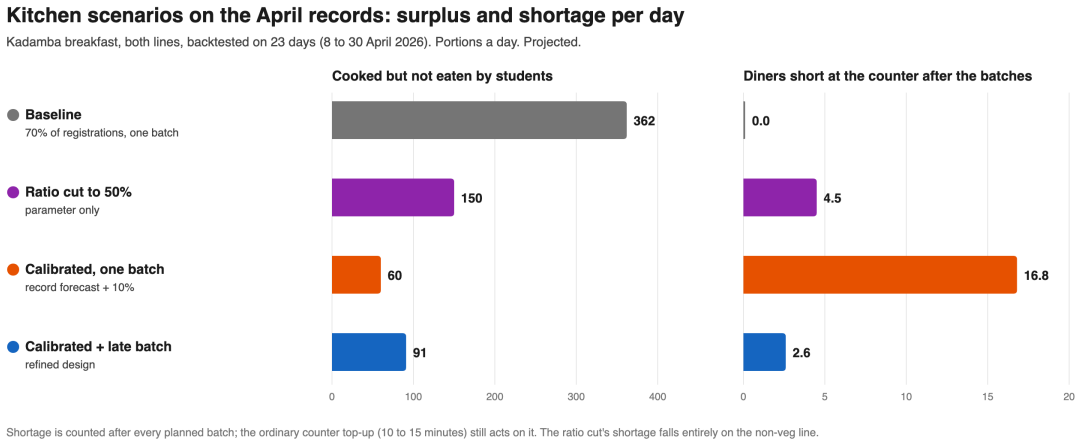


Figure 5. Kitchen rules backtested on 23 April days, portions a day, both lines (projected).

Rule	Cooked a day	Surplus a day	Short a day	Line-days short (of 46)
Baseline: 70%, one batch	759	362	0	0
G: ratio cut to 50%	542	150	4.5	7
B alone: forecast plus 10%, one batch	440	60	16.8	16
B with C: calibrated batch plus 9:00 late batch	485	91	2.6	7

Scenario 2. Projecting the day from the morning. A flat projection from the count at 8:00 assumes a quarter of diners are in by then, when only 13.5 to 22.5% are, and it under-reads by 26.9% on average (mean absolute error 34.8%, range –56% to +54%). A weekday profile at 8:00 removes most of the bias but not the noise of a small early count (over-reads by 15.6%, absolute error 28.6%). Reading at 9:00 halves the error again (over-reads by 2.6%, absolute error 13.4%, range –23% to +70%), hence the 9:00 trigger, with the card rebuilt monthly.

Scenario 3. One Sunday on the veg line. Across three backtest Sundays the baseline cooked about 563 for about 190 eaten, a surplus of about 373, which is 2.5 to 3.4 times what was eaten across ratios of 60 to 80%. The refined design would cook about 233, leaving about 45 surplus and about 2 short. As the week's largest single surplus block, Sunday veg is the natural pilot slice.

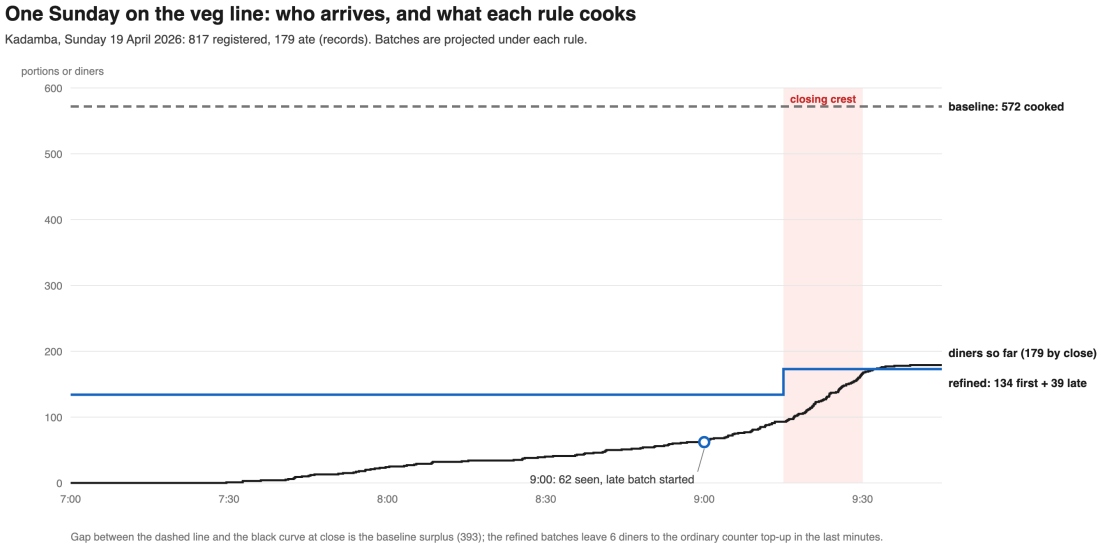


Figure 6. Kadamba veg line, Sunday 19 April 2026: recorded diners against the baseline batch and the refined first and late batches (projected).

Scenario 4. The student side and the default. In April, Kadamba students were charged for 20,221 breakfasts they did not eat (confirmed): about Rs. 7.4 lakh veg and Rs. 3.2 lakh non-veg, Rs. 10.6 lakh in all (projected). Kitchen rules leave that unchanged. A skip that returns nothing changes nothing, which is why Skip Meal fails today. A skip that removes the charge is the smallest rule change that returns money, and it scales from Sunday to every day. A shorter lead with the cap of five kept can remove at most 26.5% of no-shows (1,072 students times 5 against 20,221), which shows the cap limits any exit however it is framed.

Scenario (April charges for uneaten breakfasts)	Charges	Change
Baseline, or a skip that returns nothing	Rs. 10.58 lakh	None
D: Sunday skip, 25 to 50% uptake	Rs. 9.78 to Rs. 10.18 lakh	Rs. 0.40 to Rs. 0.80 lakh less
Sunday opt-in, turnout 60 to 85%	Rs. 9.05 to Rs. 9.35 lakh	Rs. 1.23 to Rs. 1.53 lakh less
Every-day skip, 25 to 50% uptake	Rs. 5.29 to Rs. 7.94 lakh	Rs. 2.65 to Rs. 5.29 lakh less
F (held)	None	Rs. 10.58 lakh off students; the bearer depends on the plate basis

A week of Sunday skips at 10, 25, 50 and 75% uptake returns Rs. 4,010, Rs. 10,025, Rs. 20,050 and Rs. 30,075 and removes 78, 194, 389 and 583 registered plates; the every-day skip returns Rs. 24,692, Rs. 61,729, Rs. 1.23 lakh and Rs. 1.85 lakh and removes 472, 1,180, 2,359 and 3,539 plates. Fewer registered plates cost the vendor only if it is paid on them (unconfirmed). If the kitchen keeps its ratio, opt-in starves the crest, since the ratio assumes 30% of bookings go uneaten: 70% of a confirmed count leaves each line about 23 short on Sundays at 72% opt-in turnout and about 65 short at 85%, nearly all after 9:15. A skip is less exposed. A kitchen keeping a 70% ratio stays safe on Sundays down to 82.3% uptake on veg and 67.2% on non-veg, and every day down to 66.1% and 38.0%. At 60% the Sunday thresholds fall to 72.5% and 49.0%, and at 80% they rise to 89.7% and 80.9%. A Sunday skip is therefore unlikely to short the crest below about two-thirds uptake on non-veg, even if a quarter of skippers still eat. An every-day skip at 50% uptake runs non-veg short on 7 of 30 days, about 1.4 a day, rising to 3.0 if a tenth of skippers still eat and 8.9 if a quarter do. The Sunday skip can go on its own fixed date, and the every-day skip waits until the kitchen reads eaters on every line.

Scenario 5. Yuktāhār's belief against its register. Across nine legible September breakfasts Yuktāhār registered 228 to 325 and served 41 to 105 (confirmed). Planning on the site head's ”about 50 percent” would overcook by about 63 portions a day, with uneaten charges near Rs. 10,500 a day at Rs. 53 (both projected). The register removes the gap (Sundays 18 to 24%, Tuesdays 31 to 40%). Yuktāhār's lever here is a change of belief and needs no new routine.

Scenario 6. Sensitivity and stress. Each condition is an assumed shock to Kadamba's recorded arrivals, unwarned unless stated. ”Waiting” diners arrive after one batch runs out and before the next lands. The baseline surplus is 362 a day throughout, except in the exam and favourite-item rows.

Condition	Eaten	Refined surplus	Short	Waiting
Normal (backtest)	397	91	2.6	3.4
Crest 5 points lighter	397	120	0.5	5.7
Crest 5 points heavier	397	69	7.7	1.1
Crest 10 points heavier	397	52	12.6	0.3
Exam week, 25% fewer (baseline 462)	297	123	0	0
The same, with calendar note	297	67	2.1	2.5
Favourite item, 25% more (baseline 264)	496	89	9.1	14.2
Rule change, 10% fewer (baseline 402)	357	100	1.4	0.7
Rule change, 10% more (baseline 323)	436	89	5.6	6.5
Gas shortage, 30-minute batches	397	91	2.6	25.3
Failed 9:00 late batch	397	33	43.6	32.5

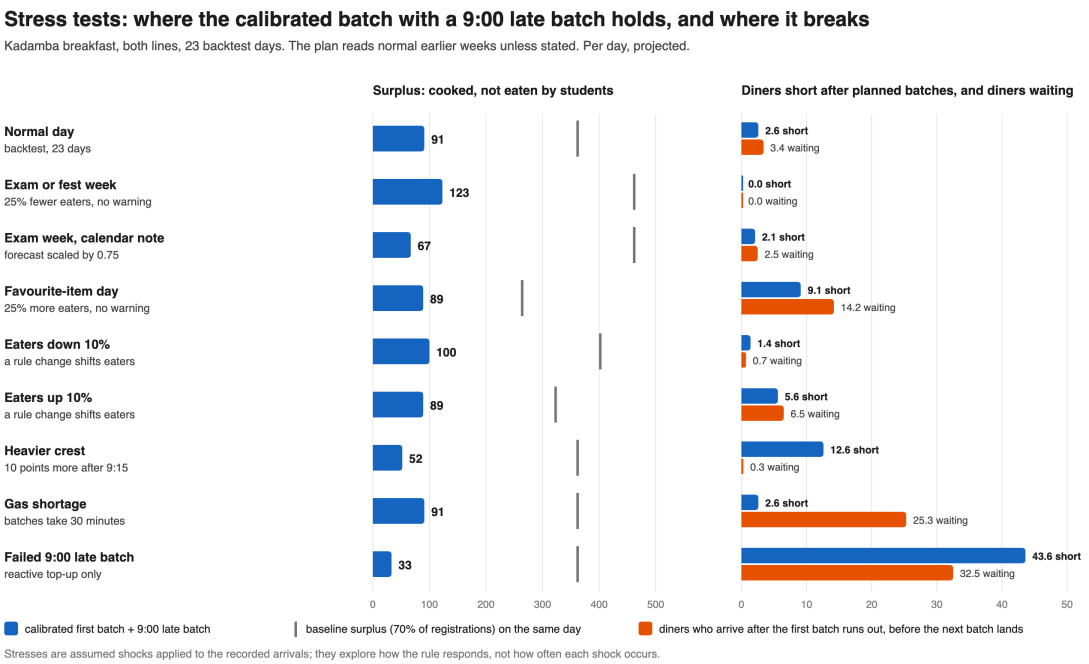


Figure 7. *The refined kitchen design under assumed shocks on the 23 backtest days, beside the baseline surplus (projected).*

From the lightest to the heaviest crest, line-days short (of 46) run 2, 7, 13 and 15. The design fails safe toward fewer eaters and loud toward more. The late batch is load-bearing: if it fails, about 44 diners are short and 33 wait, so the raw reserve is mandatory and two failed Sundays running should stop the pilot. A gas shortage hurts mainly through waiting, and on such days the plan reverts to a larger first batch. A 10% shift in eaters keeps surplus between 89 and 100, so the no-adaptation assumption is not decisive. The baseline surplus is almost pure ratio, 254 to 471 across 60 to 80%.

Scenario 7. Behaviour over time for the recommended bundle. Adoption pace and timing are assumed. The baseline holds at 362 portions a day (254 to 471) and Rs. 2.47 lakh a week throughout, and nothing physical moves in Stage 1.

Week	Stage	Surplus a day	Uneaten charges a week
0 to 4	1: routes, tests, joint reading at week 4	362 (254 to 471)	Rs. 2.47 lakh
6	2: Sunday veg pilot; skip proposed	341 (236 to 446)	Rs. 2.47 lakh
8	2	320 (219 to 421)	Rs. 2.47 lakh
10	3: roll-out; Sunday skip in force	262 (182 to 346)	Rs. 2.37 lakh (2.27 to 2.37)
12	3	205 (144 to 271)	Rs. 2.37 lakh
14	3; every-day skip decision	148 (107 to 195)	Rs. 2.37 lakh
16 to 20	Full adoption	91 (69 to 120)	Rs. 2.37 lakh; Rs. 1.23 to Rs. 1.85 lakh with the every-day skip

Surplus falls over roughly three months, perhaps optimistically, since Yuktāhār's own routine took about four (August to December, single-sourced). The charge moves only when a rule does. A kitchen improving under a flat charge line signals a quick fix standing in for the fundamental one.

Scenario 8. How each scenario moves every indicator. Figures are projected; directions given without figures are reasoned (assumed).

Indicator (today)	Quick fix only	Bundle, Sunday skip	With every-day skip	G	F (held)
Uneaten charges a week (Rs. 2.47 lakh)	Unchanged	Rs. 2.27 to 2.37 lakh	Rs. 1.23 to 1.85 lakh	Unchanged	None; moves per plate basis

Indicator (today)	Quick fix only	Bundle, Sunday skip	With every-day skip	G	F (held)
First-batch surplus a day (254 to 471)	69 to 120	69 to 120	69 to 120	150	Falls only if the kitchen reads eaters
Diners short a day (0)	0.5 to 7.7; 44 if the late batch fails	As quick fix	Non-veg short some days unless calibrated	4.5, non-veg	Unchanged
Vendor, if paid on registered plates	Food cost falls	194 to 389 fewer plates a week	1,180 to 2,359 fewer	Food cost falls	Follows served plates

Every kitchen scenario except G names the staff allowance and reheats less. Supervisor workload rises at first, and the share arriving after 9:30 may creep up within its stop rule. The bundle adds a weekly page and a portal toggle for CDS and eases pressure on Mess Cell resale. G risks the allowance and trust. Only rule scenarios move the student cost and only kitchen scenarios move the surplus, so the bundle needs both.

2.3 Simulate Stakeholder Responses

All responses are simulated from each actor's stated position and incentives, not from actor feedback. Each response is rational given the actor's incentives, which can change.

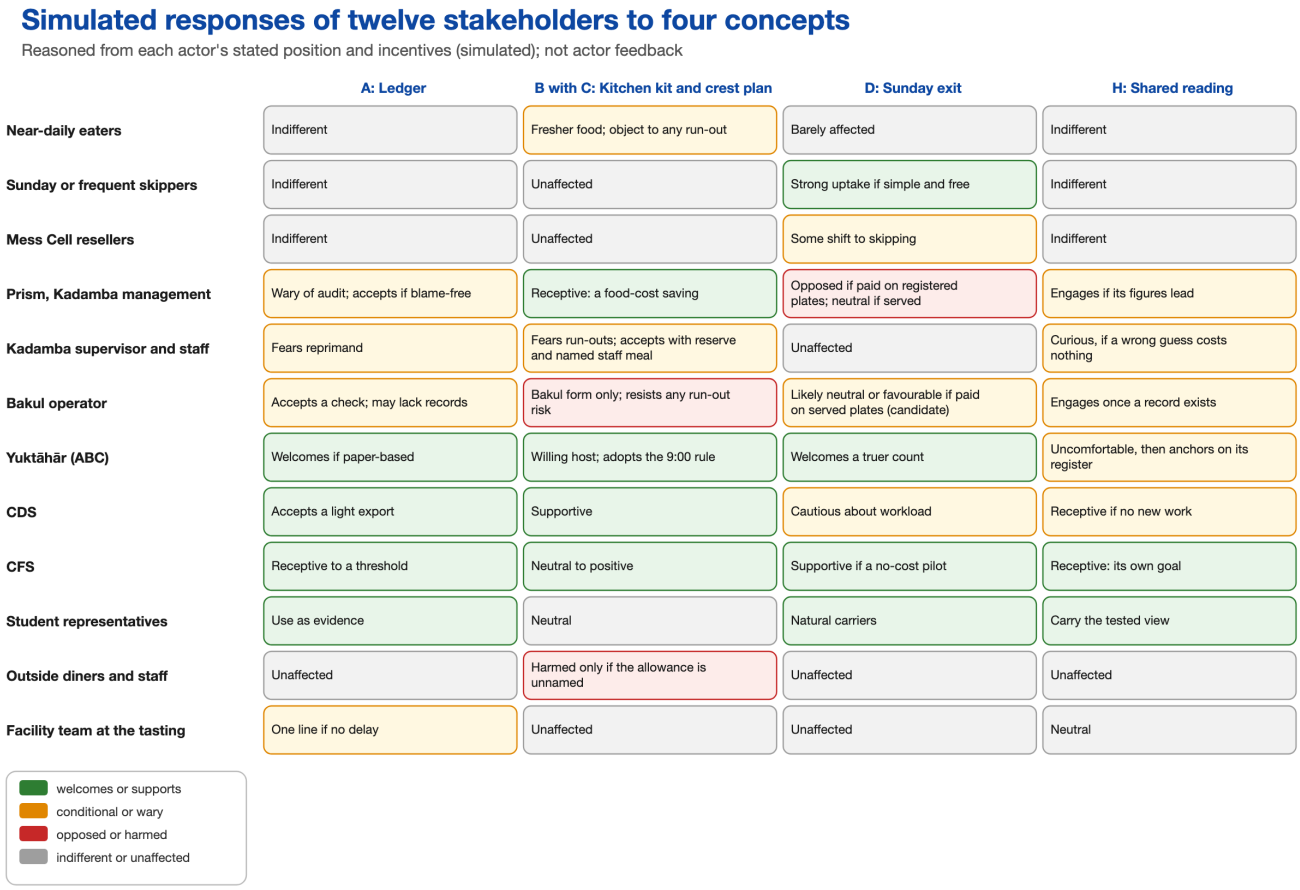


Figure 8. Simulated responses of twelve stakeholders to four concepts.

Prism needs a food-cost saving it keeps, visible run-out protection and a contract assurance that lower waste alone will not lower its rate. The supervisor needs the blame-free rule and a named staff meal. CDS needs a skip that replaces cancellation work and adds none. Bakul's floor is rational when a late call cannot land in time, so it needs a record of how often a smaller batch would have run short, and if Bakul is paid on plates served it would most likely object to shortage risk and not to a lower booking count (candidate). The simulation favours one joint reading over

separate proposals, and favours not holding the booking change behind the kitchen.

2.4 Test Interventions Against System Objectives (Deliverable 4: Evaluation Framework)

The objectives belong to the system, not to any one stakeholder: notice its own cost, cut the first batch without trading waste for run-outs, leave no one worse off, and be usable. Four gates make a concept ineligible whatever its score. G1: absorbers kept, so resale, reuse, outside-diner and staff meals, walk-ins, late service and outside venues all survive and only the over-cooked batch, reheating and the bin are targeted (P5). G2: safety kept (P4). G3: aimed at structure, neither asking students to behave better nor blaming operators, and not gated on an unknown that could reverse it (P3; LP10 held). G4: any kitchen lever paired with a route, default, goal or belief lever landing on a date independent of kitchen performance.

Concepts are then scored from 0 (worse) through 1 (no effect) to 4 (large gain in the model) against nine criteria, weighted out of 400: O1 booked against eaten, O2 cooked against eaten (never read alone), O3 crest service at every hall, O4 fundamental capability (15 each); O5 fairness, O6 food quality, O9 desirability (10 each); O7 feasibility now and O8 reversibility (5 each). The weights are the team's judgement. Food quality rose from 5 to 10 because students eat reheated surplus and worse portions are the quickest way to cut surplus, and it now includes holding time and texture after the Bakul manager reported that carried aloo paratha loses its texture (single-sourced). Feasibility and reversibility fell from 10 to 5, and cooked-against-eaten from 20 to 15, since weighting toward what a student team can do pulls every score toward the quick fix. Process checks are monitored and never scored. On O3, 3 means a residue one top-up can cover.

Concepts scored against the system objectives

Four gates first, then nine weighted criteria scored 0 to 4 (total out of 400); gate failure makes a concept ineligible whatever its score

	Gates	O1 weight 15	O2 weight 15	O3 weight 15	O4 weight 15	O5 weight 10	O6 weight 10	O9 weight 10	O7 weight 5	O8 weight 5	Score
A: Breakfast Ledger	Pass	2	2	4	4	2	1	2	4	4	68%
H: Shared Reading	Pass	2	2	4	3	3	1	2	3	4	65%
F: Billed at consumption	Fails G3 (held)	4	2	4	2	3	1	2	0	2	63%
C: Crest Plan, alone	Fails G4 alone	1	2	4	1	3	3	3	4	4	63%
D: Sunday skip, ratio kept	Pass	2	2	4	2	3	1	3	1	3	60%
E: Operator Menu Slot	Pass	1	2	4	2	2	2	3	3	4	60%
B: Kitchen Kit, alone	Fails G4 alone	1	4	1	3	2	3	1	3	4	58%
D: every-day skip	Pass	3	2	1	2	3	1	3	0	2	50%
D: Sunday opt-in (first draft)	Pass	3	3	0	2	2	2	2	1	3	50%
G: Ratio cut to 50%	Fails G1, P4	1	3	3	1	1	1	1	4	4	48%
First-draft bundle	Fails G4	2	4	3	4	3	3	3	3	3	79%
Re-sequenced bundle	Pass	2	4	3	4	3	3	3	3	3	79%

O1 booked against eaten, O2 cooked against eaten, O3 crest service every hall, O4 fundamental capability, O5 fairness, O6 food quality, O9 desirability, O7 feasibility now, O8 reversibility. Scores are model results (projected); weights are the team's judgement.

Figure 9. *Concepts scored against the gates and the nine weighted criteria.*

A leads because it builds the route every other lever needs, though alone it moves no food or money. H acts deepest, but discomfort at seeing one's own figure caps its desirability. B alone has the largest physical effect (surplus from 362 to 60) and the worst crest result (16.8 short), falling from second place to seventh. The every-day and opt-in forms of D move Rs. 2.65 to Rs. 5.29 lakh and Rs. 1.23 to Rs. 1.53 lakh a month but fail the crest test. G fails the gates outright and still leaves 150 surplus portions. The two bundles have the same parts and the same score, since a weighted score cannot see sequencing and the gate can. The first-draft bundle made the Sunday skip wait for

the kitchen to prove itself, a small case of shifting the burden. Scenario 4 shows the wait is unnecessary, so the re-sequenced bundle, which passes G4, is the one recommended.

2.5 Identify Unintended Consequences

Several second-order effects are the loops pushing back.

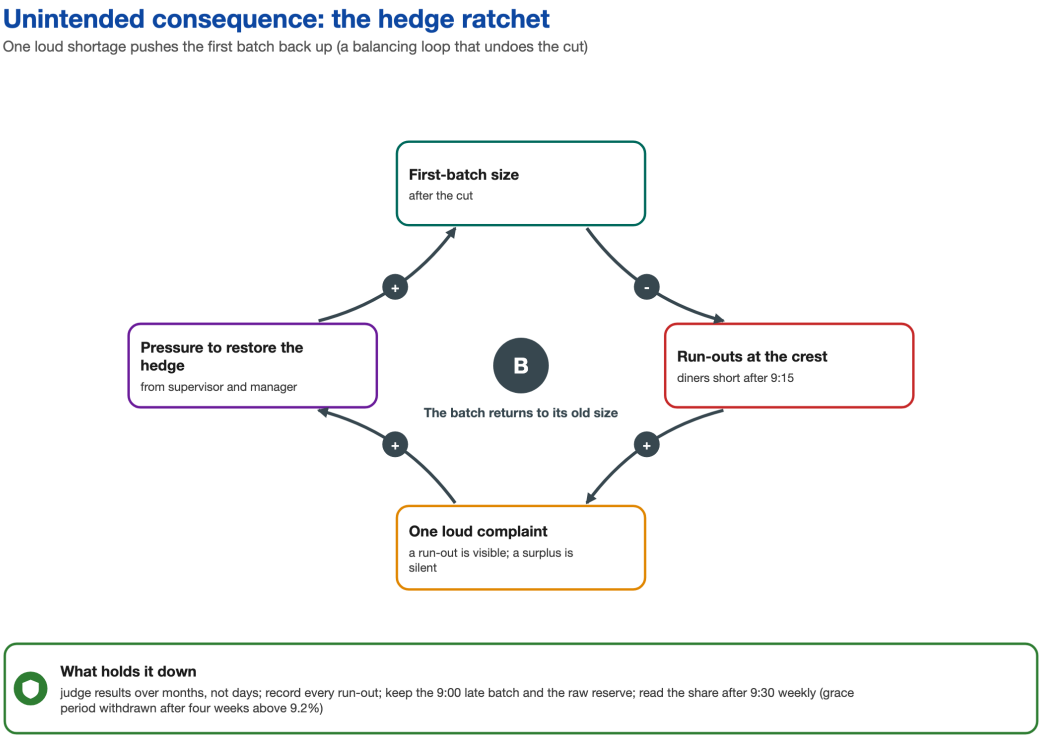


Figure 10. The hedge ratchet: one loud shortage restores the cut batch, and what holds it down.

Intervention	Consequence	Loop	Mitigation
B, C	Food taken from outside diners and staff	L9 (candidate)	Named allowance, recorded as served
B	Crest run-outs on heavy days; one loud shortage restores the hedge	L14 reversed, L12; the ratchet	Late batch, 1.5 cover, reserve; judged over months; every run-out recorded
B, A	A better kitchen hides the gap upstream	L17 (candidate)	The ledger carries the charge; the kitchen is labelled the quick fix
C	Late batch lands in the busiest quarter hour	L12	Live or steamed quick items; walk-through first
B	Exams and events break the forecast	L7	Calendar note
D	Resale buyers lose Sunday slots; opt-in with a fixed ratio under-cooks the crest	L1; L13, L14	Accepted, no market closed; the skip keeps the default
B, C	Less reheating changes what students get	L13 (open)	Recorded on the sheet
B, C at Bakul	A Kadamba rule trims a batch no top-up can replace; a bigger reserve softens held items (candidate)	L12 with delay; L13 holding (open)	No Kadamba rule applied; floor kept; reserve sized from Bakul's own arrivals; texture under O6
A	CDS workload rises	Governance	Paper-first; tested time budget
A, D	Two people on record as head of CDS	L10	Send by role

Adaptations that would push back, with likelihoods assumed:

Adaptation	Loop	Likelihood	Early warning	Mitigation
Vendor raises its rate at renegotiation	L8	Medium on registered plates, low on served	Rate proposals	Ask the plate basis first; show food-cost saving beside lost plates

Adaptation	Loop	Likelihood	Early warning	Mitigation
Skip, then eat on another's scan	L1	Medium Sundays, higher daily	Eaten per registration rising	Walk-in Rs. 81 against Rs. 48 (confirmed); daily step behind its own rule
Smaller portions, hidden leftovers, reheated stock	L11 goal; L13	Medium	Grams per plate; ”reuse” rising	Never read alone (P7); store issues reconciled
”Ate” gamed through hand-logged scans	L10	Low to medium	Hand-logged share rising	Own line; a rise triggers a scanner check, not a staff check
Allowance quietly cut	L9	Medium, least visible	Allowance below plan	A missed meal is a stop condition
CDS re-tightens cancellations	December 2024 precedent	Low to medium	Proposals to cap skips	Skip replaces cancellation work; CDS in from week 4
Censoring spiral lowers forecasts	Candidate chain on L11	Medium	Repeat weekday run-outs	Demand read as count at run-out divided by card share; two run-outs raise the forecast
Ledger read as an audit	Accidental adversaries	Medium at Kadamba	Round numbers; ”left over” always small	Blame-free rule; review, never deduction
Hedge quietly restored	L14 ratchet	Medium to high	Effective ratio drifting to 70%	Weekly ratio line
Grace period softens the close	L15 (candidate)	Medium	Share after 9:30 above 9.2%	Withdraw the grace after four weeks; the late batch stays

2.6 Refine the Intervention

Tests, stress runs, consequences and the Bakul evidence changed the first concepts in twelve ways.

First concept	What changed it	Refined version
B alone	16.8 short a day; 45 short at 9:24 one Sunday	B with C: batch for diners before 9:15, late batch at 9:00
C projecting at 8:00	Error 35% flat, 29% by profile at 8:00, 13% at 9:00	Projection at 9:00
10% first-batch buffer	About 18 diners a day left waiting	1.5 cover: 3 waiting, 2.6 short, 91 surplus
One C for every hall	Bakul's rush, phoned top-up and held reserve	C per hall; Bakul's reserve once records exist
Kitchen bundle as the whole design	Shifting the burden	Labelled the quick fix everywhere
Sunday skip after the kitchen reads eaters	Scenario 4: safe below two-thirds uptake	Fixed date, week 6; every-day skip on its own rule
Sunday opt-in	23 to 65 short; Rs. 81 for the forgetful	Evening-before skip; opt-in later on the path
Any skip	Returning nothing, it goes unused (L2)	Must remove the charge and lower the count
A without the charge	Surplus read as solved	Weekly charge line
B without allowance	G1	Named allowance on every card
A as a dashboard	Yuktāhār dropped a paid ERP system (single-sourced)	Paper-first A4 ledger
Records alone	Records cannot show use	Planned co-design tests

The late-batch design was chosen from a grid of start time, cover and buffer (projected). Starting at 8:45 with cover 1.1 and a 10% buffer gives 77 surplus, 10.6 short and 20.0 waiting; 8:45 with cover 1.5 and 20% gives 116, 4.9 and 6.2. Starting at 9:00 gives, for cover 1.1 and 10%, 67 surplus, 8.6 short and 17.7 waiting; for 1.3 and 10%, 75, 6.5 and 7.4; for 1.5 and 10%, 91, 2.6 and 3.4; and for 1.5 and 20%, 118, 0.8 and 3.4. The choice is the least surplus that keeps waiting under five a day, which is 9:00 with cover 1.5 and a 10% buffer. A daily cap at 1.5 times the forecast would save 21 more portions for 2.7 more short, and the pilot can decide.

2.7 Deliverable 5: Final System Design Recommendation

The Recommended Intervention

Route and read the gap together, open the Sunday exit on a fixed date, and calibrate the kitchen to buy time. The kitchen bundle runs alongside, labelled the quick fix wherever it is reported. Stage 1 (weeks 0 to 4) builds

the ledger with its charge line, runs the co-design tests, holds the joint reading at week 4 and takes the goal and indicators to CFS and CDS, while the cards and sheet are drafted with the Kadamba supervisor. Stage 2 (weeks 5 to 8) proposes the Sunday skip to the rule owner at week 6, through student representatives and with the plate-basis question, while the Kadamba Sunday veg pilot runs the card, allowance, 9:00 batch and sheet and Yuktāhār re-anchors. Stage 3 (weeks 9 to 20) puts the skip in force from about week 10 if adopted, takes the every-day decision at week 14, hands over from week 16 and ends at week 20. Billing and payment basis (LP10) and an outcome owner (LP9) are held until their facts are known.

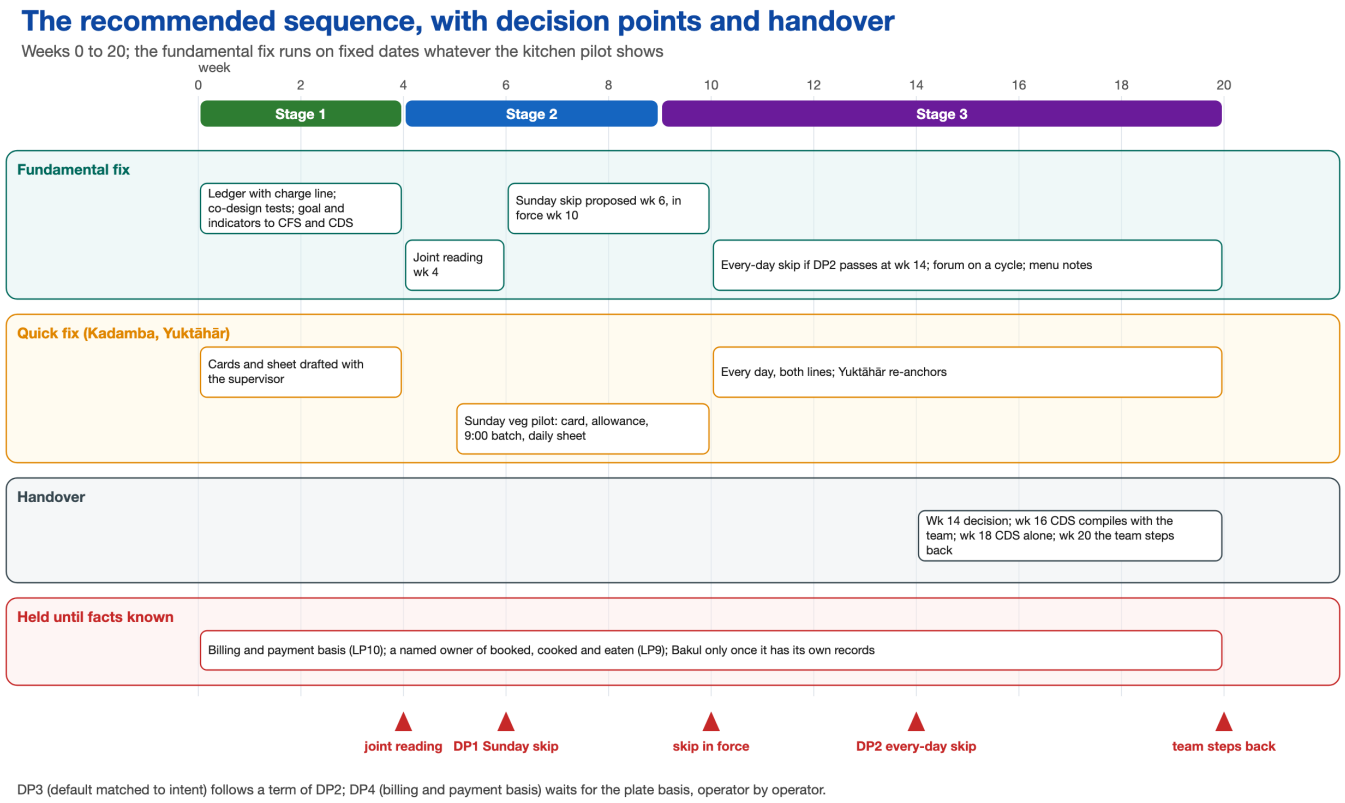


Figure 11. *The recommended sequence, with decision points and handover.*

Bakul, a later hall outside this pilot, is a third regime with no records, and the Kadamba model does not transfer. The recommendation extends there only once its scan export gives weekday turnout and arrivals, the call-to-counter lead time is measured, an item-level record backs the weekly review, any kitchen lever cuts waste without raising shortage risk, and, for the Sunday exit, its payment basis is confirmed.

The Architecture Path

The path moves in reversible steps from a default that must be escaped toward a booking that matches intent.

Decision point	When	Rule	If the evidence goes the other way
DP1: Sunday skip	Week 6, fixed; in force about week 10	None; goes on this date	If declined, the Sunday charge line goes to the escalation forum
DP2: Every-day skip	Week 14; from about week 16	Sunday uptake at least 25%; no Sunday over 10 short; eaten per remaining registration rising a few points; calibrated batch on both lines	Uptake under 10%: retest the wording. Shortage: that line reads eaters first
DP3: Default matched to intent	After a term of DP2	Opt-in on the lowest-turnout days; no automatic registration off campus; skip used by over half of no-shows; never costs a hall slot	Low uptake and a stable charge: keep skip and default

Decision point	When	Rule	If the evidence goes the other way
DP4: Billing and payment	Once the plate basis is known	Billing at consumption, or payment on served plates; basis known operator by operator; resale reliance checked	If registered plates and no rate agreement, billing stays and the skip carries students

Theory of Change

The bundle acts on the place-3 goal, shifting the working measure from registration count to cooked-against-eaten, an aim already on the institution's own poster.

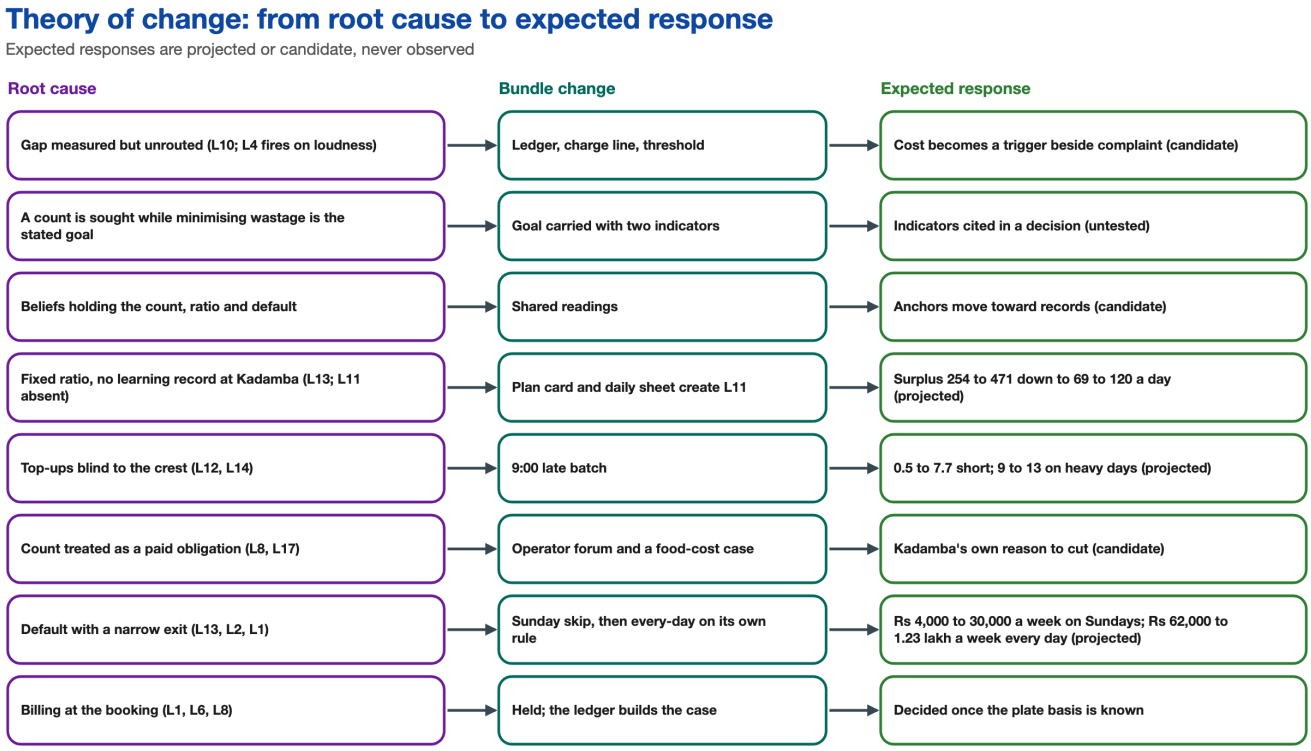


Figure 12. From root cause to bundle change to expected response (projected or candidate).

Interventions Across the Iceberg

The daily sheet records events (batches, run-outs, leftovers), marking run-out days as censored by run-out time per line. Patterns come from the monthly arrival card and the ledger's behaviour-over-time lines: weekly surplus, shortage and charge, the share after 9:30 and the effective ratio. Structures (the ledger route, plan card, skip and stated goal) are tracked by whether the ledger reaches the rule owner, by skip uptake and by the two indicators. The mental models are ”registration is demand,” ”breakfast is about 50 percent,” ”cooking to registrations is what we are paid for,” ”we cannot nullify it” and ”they are catching us.” The shared readings address each holder, and the Bakul manager's beliefs stay open questions. A shift shows when the site head's figure matches his own register.

Pilot Design and Monitoring

Element	Specification
Where and when	Kadamba veg line, Sunday breakfast: four Sundays in Stage 2, continued in Stage 3, judged at week 16; Yuktāhār for re-anchoring
Convening	A joint reading at week 4 agrees the definition of ”ate” and hears the Sunday proposal; repeated at week 12
Runs and agrees	The Kadamba supervisor runs it and the team fills the ledger; Prism's operations manager, ABC Hospitality Services, the facility team's parent body and the rule owner agree

Element	Specification
Result measures	Uneaten charges; surplus; run-outs and diners short; grams, taste and texture; holding time; share from a fresh or late batch; allowance meals
Process checks	Sheet nine days in ten; ledger weekly; late batch by 9:05; sessions on date (never reported as results)
Comparison	Sunday non-veg, Saturday veg and earlier Sundays
Success	Sunday veg surplus from about 370 to under 100 (model about 45); no Sunday over 10 short; run-outs no more frequent than on comparison days; every allowance meal served; grams and taste held; safety recorded; no sheet-based reprimand; charge line reported weekly
Stop	Shortage beyond one top-up two Sundays running; late batch failing twice; any allowance meal missed; grams below standard; any figure used against staff; consent withdrawn
Would change the plan	The base of the waste figure; whether Skip Meal lowers the count; the plate basis; co-design results

To keep the team from becoming the intervenor the system depends on, the CDS office compiles the page by role, operators' data roles fill their sheets, the escalation forum receives the threshold, and a named outcome owner (LP9) takes over once the head of CDS is known. CDS compiles with the team from week 16 and alone from week 18. The team steps back at week 20, once two pages and one forum have run without it, and each archetype is then checked monthly for at least six months.

Archetype	Indicator after handover	Signal of return
Shifting the burden: reliance on the quick fix	Kitchen-led fall in surplus against fall in charges	Surplus falling, charge flat for three months
Shifting the burden: fundamental progress	Decision points on date; indicators cited; uptake	A decision point missed twice
Fixes that fail	Effective ratio, weekly per line	Drift toward 70% for four weeks
Soft close (L15)	Share after 9:30	Above 9.2% for four weeks
Accidental adversaries	Completion; round numbers; hand-logged share	Completion under nine in ten for two weeks

Each waste indicator is read beside the run-out indicator for the same hall and line. Some results improve before they worsen (a bad crest can bring the hedge back months later), and others worsen first (supervisor workload in Stage 2, the charge flat until week 10). Each expected shape is written on the ledger beforehand.

Who Must Act

Of thirteen actions the student team can complete two alone: building the ledger, cards and sheet, and running the co-design tests. It can convene or support four that others decide: the joint and shared readings (CDS convenes), taking Yuktāhār's routine to Kadamba, the calibrated and 9:00 batches (Prism's supervisor decides) and Yuktāhār's re-anchoring. It can carry six forward as evidenced proposals: the tasting line (facility team, parent body unknown), a cost threshold (Student Council, Parliament, CDS Committee), the stated goal (CFS and CDS), the Sunday exit (rule owner, Mess Office or CDS, unconfirmed), the billing or payment basis (only once the plate basis is answered) and an owner of booked, cooked and eaten (CFS and CDS). One it must complete outright: the handover, by week 20.

Before and After

First-batch surplus falls from about 254 to 471 portions a day to about 69 to 120 over roughly four months. The weekly uneaten charge falls from about Rs. 2.47 lakh to Rs. 2.27 to Rs. 2.37 lakh with the Sunday skip, and to Rs. 1.23 to Rs. 1.85 lakh if the every-day skip follows (all projected). The kitchen change is large and the student-side change small until the rule moves, which matches the diagnosis. Kitchen levers cut surplus and run-outs, and only a rule moves the booking. The design keeps both in view so the first is never mistaken for the second.

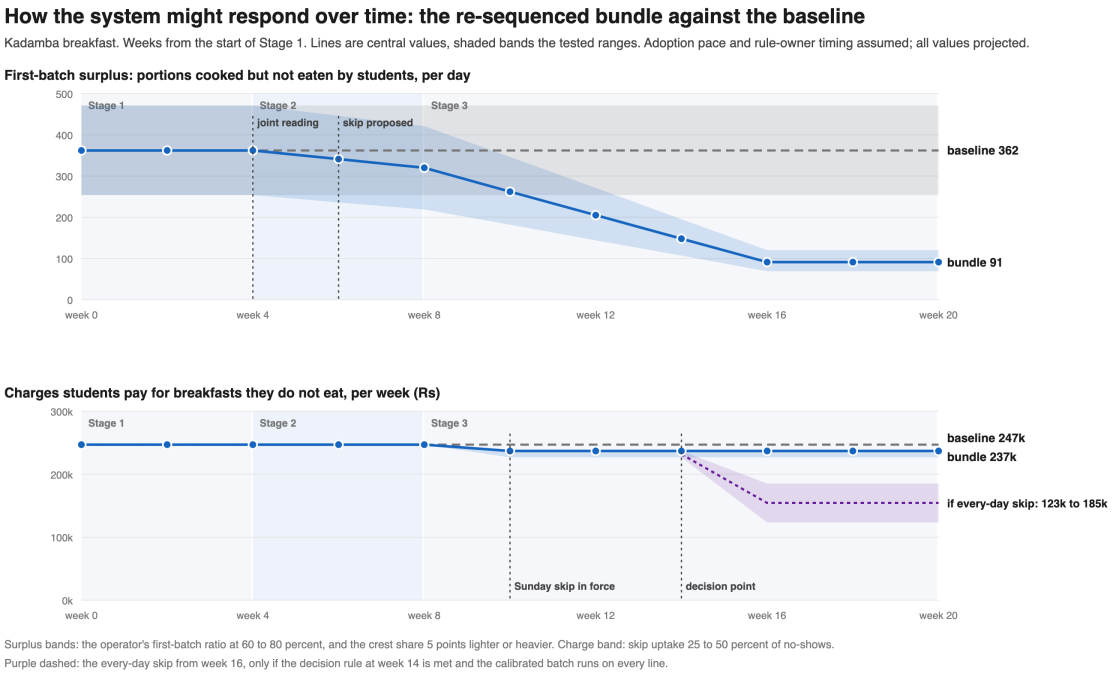


Figure 13. *How Kadamba breakfast might respond over time under the bundle, against the baseline (projected).*

3 Key Findings

1. The system has its measurements and no routes for them. The enabling lever (place 6 of 12) routes existing records to the kitchen and the rule owners, and a student team can prototype it now as a paper ledger.
2. The loops form a shifting-the-burden structure. Kitchen levers are the quick fix. Booking, goal, belief and route levers are the fundamental fix, which the recommendation runs on fixed dates whatever the kitchen pilot shows.
3. The deepest levers are the beliefs that let the registration count stand in for demand (place 2) and the system's working goal (place 3). The deepest rule lever not gated on an unknown is the breakfast default and its exit (place 5), provided no skip ever costs a student a hall slot.
4. Cutting Kadamba's ratio to 50% fails on both counts, and a calibrated first batch alone moves the cost into the crest. Adding a 9:00 late batch brings surplus down with a residue the ordinary top-up covers (projected). The design fails loud when eaters exceed the forecast, so it depends on the raw reserve and the stop rules.
5. Kitchen levers leave the student cost untouched, and only a rule that returns something moves it. A Sunday skip is unlikely to run the crest short below about two-thirds uptake on non-veg, so it can start on a fixed date now. The every-day skip waits on its own decision rule.
6. Kadamba and Yuktāhār show that a learning practice is possible under the same rules. The halls differ in five other ways yet leave a similar breakfast gap, so the cause stays a candidate reading. Targets are anchored outside each hall's past and paired with portion, quality and run-out measures, and no lever may cut waste by raising shortage risk.
7. Every modelled figure here is Kadamba's. Bakul, the third regime, has no records and needs its own scans, a measured top-up lead time and an item-level record before any lever can be sized. Sunday breakfast on Kadamba's veg line is the first test slice.

Outstanding Evidence

Still open: what a plate counts for in each operator's monthly payment; whether Skip Meal lowers the count Kadamba cooks from; the base of Kadamba's 5 to 10% waste figure; who heads CDS; whether a late batch can be cooked at 9:00 without delaying crest service; the real first batch per line and weekday and whether the supervisor already cooks in stages; Sunday skip uptake (assumed 25 to 50%); which registration regime applied to Yuktāhār's June

breakfasts; and Bakul's scans, top-up lead time and any written record. Every co-design test is planned and none has yet run.